

Facets of Brownian Motion

Srinjena Das

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Chapter 1

Introduction to Brownian motion

We are aware of one-dimensional stochastic processes. It can be generalized to a d-dimensional stochastic process as follows:

Definition 1.0.1 (d-dimensional stochastic process). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. A d -dimensional stochastic process indexed by $I \subset [0, \infty)$ is a family of random variables $X_t : \Omega \rightarrow \mathbb{R}^d$. It is denoted by $X = (X_t)_{t \in I}$, where I is the index set and $\mathbb{R}^d$ is the state space.

Note: Throughout, we will be considering $(\Omega, \mathcal{A}, \mathbb{P})$ to be our probability space.

Now, let us start with defining a Brownian motion: A Brownian motion is a d-dimensional stochastic process which is continuous w.r.t. the time variable (i.e. the stochastic process follows a continuous trajectory) and also satisfies certain other properties. The formal definition of it is given below.

Definition 1.0.2 (Brownian motion). A d-dimensional Brownian motion $B = (B_t)_{t \geq 0}$ is a stochastic process on $(\Omega, \mathcal{A}, \mathbb{P})$, indexed by $[0, \infty)$, taking values in $\mathbb{R}^d$ such that

1. $B_0(\omega) = 0$ for almost all ω .
2. For all $n \geq 0$, $0 = t_0 < t_1 < t_2 < \dots < t_n < \infty$, $B_{t_n} - B_{t_{n-1}}, \dots, B_{t_1} - B_{t_0}$ are independent.
3. $B_t - B_s \sim B_{t+h} - B_{s+h}$ such that $s + h \geq 0$.
4. $B_t - B_s \sim N(0, t - s)^{\otimes d}$, $N(0, t)(dx) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right) dx$.
5. For all ω , $t \mapsto B_t(\omega)$, $t \in [0, 1]$ is a continuous function.

We will use BM^d as a shorthand notation for d-dimensional Brownian motion.

By definition, a Brownian motion is a stochastic process starting at the origin, so $(B_t + x)_{t \geq 0}$ defines a stochastic process starting at $x \in \mathbb{R}^d$. As per requirement, our indexing set can differ from $[0, \infty)$ to $[0, T]$ or $[0, T)$. We shall use the notations $B_t(\omega)$, $B(t)$, $B_t(\omega)$, B_t for a Brownian motion as per interest.

1.1 Gaussian process and Brownian motion

We would like to first observe some relationship between Gaussian processes and Brownian motion. To this end, we first recall the following definition.

Definition 1.1.1 (Gaussian random variable). Let Γ be a one-dimensional random variable on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. We say that Γ is Gaussian if the characteristic function of Γ is of the form

$$\mathbb{E} e^{i\xi\Gamma} = e^{im\xi - \frac{1}{2}\sigma^2\xi^2}$$

for all $\xi \in \mathbb{R}$, where $m = \mathbb{E} \Gamma$, and $\sigma^2 = \mathbb{V} \Gamma$.

The following proposition tells that a Brownian motion is a Gaussian random variable with mean 0 and variance t .

Proposition 1.1.2. *Let $(B_t)_{t \geq 0}$ be a one-dimensional Brownian motion. Then each B_t is a Gaussian random variable with mean 0 and variance t . That is, the characteristic function of B_t is*

$$\mathbb{E}e^{i\xi B_t} = e^{-\frac{1}{2}t\xi^2}.$$

The following corollary then tells us about the moments of Brownian motion, and is obtained as a corollary to the above result.

Corollary 1.1.3. *Let $(B_t)_{t \geq 0}$ be a one-dimensional Brownian motion. Then for any $t \geq 0$, exponential moments of all orders of B_t exists. That is,*

$$\mathbb{E}e^{\zeta B_t} = e^{t\zeta^2/2}$$

for all $\zeta \in \mathbb{C}$. □

The moments of a Brownian motion can be calculated quite easily by writing the Taylor series and differentiating it.

Lemma 1.1.4. *Let $(B_t)_{t \geq 0}$ be a Brownian motion. Then,*

1. $\mathbb{E}(B_t^{2k+1}) = 0$ for all $k = 0, 1, \dots$.
2. $\mathbb{E}(B_t^{2k}) = t^k \cdot (2k-1) \cdot \dots \cdot 3 \cdot 1$ for all $k = 0, 1, \dots$.

In particular, we have $\mathbb{E}(B_t) = \mathbb{E}(B_t^3) = 0$, $\mathbb{V}(B_t) = \mathbb{E}(B_t^2) = t$ and $\mathbb{E}(B_t^4) = 3t^2$. □

Remark 1.1.5 (Covariance of a Brownian motion). Let (B_t) be a Brownian motion. Since, the expectation of a BM¹ is 0, covariance of B_s and B_t is given by $\mathbb{E}(B_s B_t)$. If $s \leq t$, we observe by definition of Brownian motion that

$$\mathbb{E}(B_s B_t) = \mathbb{E}(B_s(B_t - B_s)) + \mathbb{E}(B_s^2) = s = \min\{s, t\}.$$

This will come in handy later on.

Our main goal now is to show that Brownian motion is equivalent to certain type of Gaussian processes. Before that we recall what is a Gaussian process.

Definition 1.1.6 (Gaussian process). A one-dimensional stochastic process $(X_t)_{t \geq 0}$ is a Gaussian process if for any $n \geq 1$ and any $0 \leq t_1 < t_2 < \dots < t_n$, we have that the vector $\Gamma = (X_{t_1}, \dots, X_{t_n})$ is a Gaussian random vector.

The following characterization is the main theorem in this regard.

Theorem 1.1.7 (Characterization of BM¹). *Let $(B_t)_{t \geq 0}$ be a one-dimensional stochastic process. Then the following are equivalent:*

1. $(B_t)_{t \geq 0}$ is a one dimensional Brownian motion.
2. $(B_t)_{t \geq 0}$ is a Gaussian process, such that for all $n \in \mathbb{N}$, $\Gamma = (B_{t_1}, \dots, B_{t_n})$ for $0 \leq t_1 < t_2 < \dots < t_n < \infty$ is a Gaussian random vector with mean 0, covariance matrix $C = (t_i \wedge t_j)_{i,j=1,\dots,n}$, and continuous sample paths.

1.2 Brownian motion in $\mathbb{R}^d$

We will now study Brownian motion in d -dimensional Euclidean space. Our main result is that this is equivalent to a d -tuple of independent BM¹ processes. To this end, we first need the following small result.

Lemma 1.2.1. *Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process. Then the following are equivalent:*

1. $(X_t)_{t \geq 0}$ satisfies axioms 1-4 of Definition 1.0.2.
2. For all $n \geq 0$, $0 = t_0 < t_1 < \dots < t_n$ and $\xi_0, \dots, \xi_n \in \mathbb{R}^d$, we have

$$\mathbb{E} \left(\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right) = \exp \left(-\frac{1}{2} \sum_{j=1}^n |\xi_j|^2 (t_j - t_{j-1}) \right).$$

Proof. (1. $\Rightarrow$ 2.) We expand LHS as follows. First, observe that by expectation of Gaussian random variable we have

$$\mathbb{E} \left(\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right) = \exp(i \langle \xi_0, X_{t_0} \rangle) \cdot E \left(\prod_{j=1}^n \exp(i \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle) \right).$$

By axiom 1, we deduce that $\exp(i \langle \xi_0, X_{t_0} \rangle) = 0$. By axiom 2, the above reduces to

$$\mathbb{E} \left(\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right) = \prod_{j=1}^n \mathbb{E} \left(\exp(i \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle) \right).$$

By axiom 3, we get that

$$\prod_{j=1}^n \mathbb{E} \left(\exp(i \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle) \right) = \prod_{j=1}^n \mathbb{E} \left(\exp(i \langle \xi_j, X_{t_j - t_{j-1}} \rangle) \right).$$

Finally, by axiom 4, we have

$$\prod_{j=1}^n \mathbb{E} \left(\exp(i \langle \xi_j, X_{t_j - t_{j-1}} \rangle) \right) = \prod_{j=1}^n \exp \left(-\frac{1}{2} (t_j - t_{j-1}) |\xi_j|^2 \right),$$

as required.

(2. $\Rightarrow$ 1.) By fixing $k = 0, \dots, n$, letting $\xi_j = 0$ if $j \neq k$ and using them in the equation of the hypothesis, the result is immediate. $\square$

Using the above lemma, we may prove the following important result.

Theorem 1.2.2 (Coordinate processes of BM^d). *Let $B = (B_t)_{t \geq 0}$ be a d -dimensional stochastic process. Then the following are equivalent:*

1. B is a d -dimensional Brownian motion.
2. The coordinate processes $B = (B^1, \dots, B^d)$ are each 1-dimensional independent Brownian motion.

Proof. (1. $\Rightarrow$ 2.) We first show that B^k is a BM^1 . As B is a BM^d , therefore we deduce that B satisfies the equation in Lemma 1.2.1. Let $(e_k)_{k=1,\dots,d}$ be the unit vectors of $\mathbb{R}^d$. Now fix $k = 1, \dots, d$. Taking $\xi_j = z_j e_k$ and using it in the equation, we deduce that B^j satisfies the equation in Lemma 1.2.1, 2, so by the same lemma, B^j satisfies axioms 1-4. As B has continuous sample paths, then so does B^j , as required.

We now need only show the independence of B^k . We need only show that for any $n \geq 1$ and $0 = t_0 < t_1 < \dots < t_n$, the random vectors $(B_{t_1}^k, \dots, B_{t_n}^k)$ are independent. Indeed, as we may write $B_{t_l}^k = \sum_{j=1}^l B_{t_j}^k - B_{t_{j-1}}^k$, then the above independence is a simple consequence of axioms of Brownian motion together with Lemma 1.2.1.

(2. $\Rightarrow$ 1.) Let $B_t^1, \dots, B_t^d$ be d independent BM^1 . We wish to show that the stochastic process $B = (B_t^1, \dots, B_t^d)$ is a BM^d . To this end, we first observe that continuity of sample paths follows from each B^k being a BM^1 . Hence, it suffices to check that the equation in Lemma 1.2.1, 2 is satisfied by B . Indeed, fix $n \geq 1$, $0 = t_0 < t_1 < \dots < t_n$ and $\xi_j^k \in \mathbb{R}$ for $j = 1, \dots, n$ and $k = 1, \dots, d$. Then, as B^k is a BM^1 , therefore we get the following equation for each k :

$$\mathbb{E} \left(\exp \left(i \sum_{j=1}^n \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k) \right) \right) = \exp \left(-\frac{1}{2} \sum_{j=1}^n |\xi_j^k|^2 (t_j - t_{j-1}) \right).$$

Multiplying all the k -many equations, we get

$$\mathbb{E} \left(\exp \left(i \sum_{k=1}^d \sum_{j=1}^n \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k) \right) \right) = \exp \left(-\frac{1}{2} \sum_{k=1}^d \sum_{j=1}^n |\xi_j^k|^2 (t_j - t_{j-1}) \right),$$

which is the equation for Lemma 1.2.1, 2 for B , as required. $\square$

1.3 Invariance properties of Brownian motion

Let $B = (B_t)_{t \geq 0}$ be a d -dimensional Brownian motion. We shall see that the axioms of B being a Brownian motion is preserved under various operations on sample paths. We collect all the results in the following statement below.

Theorem 1.3.1 (Invariance of BM^d). *Let $B = (B_t)_{t \geq 0}$ be a BM^d .*

1. **Reflection.** *The stochastic process $-B = (-B_t)_{t \geq 0}$ is a BM^d .*
2. **Renewal.** *Let $a > 0$ be a fixed time. Then the stochastic process $W = (W_t)_{t \geq 0}$ for $W_t = B_{t+a} - B_a$ is a BM^d .*
3. **Markov property.** *Fix $a > 0$. Let $W = (W_t)_{t \geq 0}$ for $W_t = B_{t+a} - B_a$ be the renewed BM^d . Then $(B_t)_{0 \leq t \leq a}$ and $(W_t)_{t \geq 0}$ are independent.*
4. **Time inversion.** *Fix $a > 0$. Let $W = (W_t)_{0 \leq t \leq a}$ where $W_t = B_{a-t} - B_a$. Then W is a BM^d .*
5. **Scaling.** *Fix $c > 0$. For each $t > 0$, we have that $B_{ct} \sim c^{1/2} B_t$. In particular, $(c^{-1/2} B_{ct})_{t \geq 0}$ is again a BM^d .*
6. **Projective reflection at ∞ .** *The process*

$$W_t = \begin{cases} t B_{1/t}, & \text{if } t > 0, \\ 0, & \text{if } t = 0 \end{cases}$$

is a BM^1 .

Proof. See §2.3 of cite[Schilling]. $\square$

Chapter 2

Construction of Brownian motion

Out of several different ways to construct Brownian motion, we have specified few of them in this chapter. We restrict our discussion to case of 1-dimensional motion as we have seen that any d -dimensional Brownian motion can be obtained of d independent 1-dimensional Brownian motion (Theorem 1.2.2).

2.1 The Lévy-Ciesielski construction

Let us consider the Hilbert space $L^2([0, 1])$ and a complete orthonormal system $\Phi = (\phi_n)_{n \geq 0}$. The idea behind this construction is that we will try to represent the paths $t \rightarrow B_t(\omega)$ for almost all ω as a random series wrt Φ . We consider the usual inner product structure on L^2 .

Now, we consider a family of real valued iid standard Gaussian random variables $(G_n)_{n \geq 0}$ on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$. We set

$$W_N(t) := \sum_{n=0}^{N-1} G_n \langle \mathbf{1}_{[0,t]}, \phi_n \rangle_{L^2} = \sum_{n=0}^{N-1} G_n \int_0^t \phi_n(s) ds.$$

and show that as $n \rightarrow \infty$, $W_n(t)$ will define a Brownian motion. We first start with the the following lemma.

Lemma 2.1.1. *The limit $W(t) := \lim_{N \rightarrow \infty} W_N(t)$ exists for every $t \in [0, 1]$ in $L^2(\mathbb{P})$ and the process $W(t)$ satisfies the conditions (1-4) of the definition of Brownian motion.*

To claim that $W(t)$ is indeed a Brownian motion, we still need to proof that the paths $t \rightarrow W(t, \omega)$ is almost surely continuous. We will give a proof of it using a special complete orthonormal system.

2.1.1 The Haar and Schauder systems

The Haar and Schauder systems provides us with new orthonormal basis for $L^2[0, 1]$ apart from the usual one that includes trigonometric functions.

We define the family of Haar H_{2^j+k} functions, and for $n = 0$, and $n = 2^j + k$, $j \geq 0$, $k = 0, 1, \dots, 2^j - 1$ as follows:

- $H_0(t) = 1$, and,
- $H_{2^j+k} = \begin{cases} +2^{j/2} & \text{if } t \in [\frac{k}{2^j}, \frac{2k+1}{2^{j+1}}) \\ -2^{j/2} & \text{if } t \in [\frac{2k+1}{2^{j+1}}, \frac{k+1}{2^j}) \\ 0 & \text{otherwise.} \end{cases}$

and, the family of Schauder S_{2^j+k} functions, and for $n = 0$, and $n = 2^j + k$, $j \geq 0$, $k = 0, 1, \dots, 2^j - 1$ as

- $S_0(t) = 1$, and,
- $S_{2^j+k} = \int_0^t H_{2^j+k}(s) ds$.

Clearly, $\text{supp } S_n = \text{supp } H_n$, and with very easy calculations it follows that the Haar system forms an orthonormal system in $L^2[0, 1]$. Also, by observing the graph of Schauder functions, one can see that they are tent-like functions with maximum value $\frac{1}{2}2^{-\frac{j}{2}}$ at the tip of the tent.

Now, the following theorem gives us the existence of a probability space where with the help of our Haar basis and lemma 2.1.1 we can define a Brownian motion.

Theorem 2.1.2. *There is a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a sequence of iid standard Gaussian variables $(G_n)_{n \geq 0}$ such that*

$$W(t, \omega) = \sum_{n=0}^{\infty} G_n(\omega) S_n(t) < \mathbb{1}_{[0,t]}, H_n >_{L^2}, \quad (2.1)$$

$t \in [0, 1]$ is a Brownian motion.

Proof. We consider then same probability space as we used in the definition of $W(t)$ which supports the countably many Gaussian random variables. The only thing left to proof for showing that $W(t)$ defines a Brownian motion is that the sample paths $t \rightarrow W(t, \omega)$ are continuous. To start proving this, first note that the partial sums

$$t \rightarrow W_N(t) = \sum_{n=0}^{N-1} G_n(\omega) S_n(t)$$

are continuous for all $N \geq 1$. It is enough to show that a subsequence of $(W_N(t))_{N \geq 0}$ converges uniformly to $W(t)$, as then it would be a limit of continuous functions, hence continuous. Now, we define the function

$$\Delta_j(t) := W_{2^{j+1}}(t) - W_{2^j}(t) = \sum_{k=0}^{2^j-1} G_{2^j+k} S_{2^j+k}(t)$$

Then using the orthogonality of the Schauder functions and the fact that G_n are a sequence of standard Gaussian random variables, it follows that

$$\mathbb{E} \left[\sup_{t \in [0,1]} |\Delta_j(t)|^4 \right] \leq 3 \cdot 2^{-j}, \quad (2.2)$$

for all $j \geq 1$. Now for $N > n$, using Minkowski's inequality in $L^4(\mathbb{P})$ and equation 2.2, one can show that

$$\left\| \sup_{t \in [0,1]} |W_{2^N}(t) - W_{2^n}(t)| \right\|_{L^4} \leq 3^{1/4} \sum_{j=n+1}^N 2^{-j/4} \rightarrow 0$$

as $n, N \rightarrow \infty$, and then using Fatou's lemma we have

$$\left\| \liminf_{n, N \rightarrow \infty} \sup_{t \in [0,1]} |W_{2^N}(t) - W_{2^n}(t)| \right\|_{L^4} = 0.$$

This shows that there exists a subset $\Omega_0 \subset \Omega$ such that $\mathbb{P}(\Omega_0) = 1$ such that

$$\liminf_{n, N \rightarrow \infty} \sup_{t \in [0,1]} |W_{2^N}(t) - W_{2^n}(t)| = 0,$$

for all $\omega \in \Omega_0$. By completeness of the space of all continuous functions, there is a subsequence $(W_{2^j}(t, \omega))_{j \geq 1}$ which converges uniformly in $t \in [0, 1]$ to the limiting function $W(t, \omega)$, for all $\omega \in \Omega_0$. Finally, our required one-dimensional Brownian motion is

$$\tilde{W}(t, \omega) := \begin{cases} W(t, \omega) & \text{if } \omega \in \Omega_0, \\ 0 & \text{if } \omega \notin \Omega_0 \end{cases}$$

as $\mathbb{P}(\Omega \setminus \Omega_0) = 0$ and lemma 2.1.1 remains valid for $\tilde{W}(t)$. $\square$

Now using the Brownian motion $W(t)$ we defined on $[0, 1]$, we can use the following trick to define a new Brownian motion on the entire positive real line as follows.

Corollary 2.1.3. *let $(W^k(t))_{t \in [0, 1]}$, $k \geq 0$ be independent real valued Brownian motion defined on the same probability space. Then*

$$B(t) := \begin{cases} W^0(t) & \text{if } t \in [0, 1) \\ W^0(1) + W^1(t - 1) & \text{if } t \in [1, 2) \\ \sum_{j=0}^{k-1} W^j(1) + W^k(t - k) & \text{if } t \in [k, k + 1), k \geq 2. \end{cases}$$

is a Brownian motion indexed by $t \in [0, \infty)$.

The construction that we have seen above was together given by Lévy-Ciesielski. But, to motivate ourselves on how we can think of writing our Brownian paths as a random series, we need to go back to the initial argument provided by Lévy on construction of Brownian motion via the method of interpolation.

2.1.2 Levy's original argument

Let $t_0 < t < t_1$ and $(W_t)_{t \in [0, 1]}$ be a one-dimensional Brownian motion. Define $G' = W(t) - W(t_0)$ and $G'' = W(t_1) - W(t)$. Note that by axioms of Brownian motion, both are independent Gaussian random variables with mean 0 and variance $t - t_0$ and $t_1 - t$. We wish to simulate $W(t_0)$, $W(t)$ and $W(t_1)$. To this end, we can do either of the following two:

1. determine $W(t_0)$ and then G', G'' ,
2. determine $W(t_0)$ and $W(t_1) - W(t_0)$ and then $W(t)$ by interpolation.

Let Γ be a standard Gaussian random variable mean 0 and variance t which is independent of $W(t_0)$ and $W(t_1) - W(t_0)$. We get

$$\begin{aligned} \Gamma' &= \frac{(t_1 - t_0)W(t_0) + (t - t_0)W(t_1)}{t_1 - t_0} + \sqrt{\frac{(t_1 - t)(t - t_0)}{t_1 - t_0}} \Gamma \\ &= W(t_0) + \left(\frac{t - t_0}{t_1 - t_0} \right) [W(t_1) - W(t_0)] + \sqrt{\frac{(t_1 - t)(t - t_0)}{t_1 - t_0}} \Gamma. \end{aligned}$$

Note that the above equalities follows from independence of $W(t_0)$, $W(t_1 - t_0)$ and Γ .

It follows from above that Γ' is a Gaussian random variable with the same distribution as $W(t)$ and so

$$\mathbb{P}(W(t) \in \cdot \mid W(t_0) = x_0, W(t_1) = x_1) = N(m_t, \sigma_t^2)$$

where

$$m_t = \frac{(t_1 - t)x_0 + (t - t_0)x_1}{t_1 - t_0}$$

$$\sigma_t^2 = \frac{(t_1 - t)(t - t_0)}{t_1 - t_0}.$$

Let t be the midpoint of $[t_0, t_1]$ where we keep $t_0 = 0$ and $t_1 = 1$. Assume we know $W(t_0)$ and $W(t_1)$. We now do the required interpolation.

1. Interpolate $W(0)$ and $W(1)$ linearly. As $t = 1/2$. Since we have started with $t_0 = 0$ and $t_1 = 1$, this always us to simulate the values $W\left(\frac{k}{2^j}\right)$ where $k = 0, 1, \dots, 2^j - 1, 2^j$ where $j \geq 0$. See Figure 2.1.

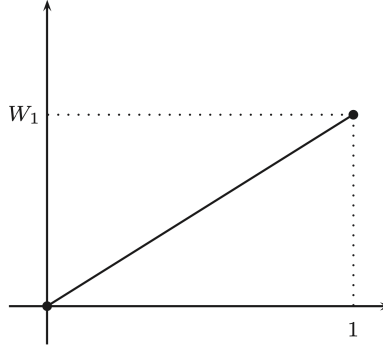


Figure 2.1: Step 1.

2. After interpolating $W(t)$ and $W(t_1)$ linearly, we consider the mid-point of this line and add an independent Gaussian random variable with mean 0 and variance $\sigma^2 = \frac{(t-t_0)(t_1-t)}{t_1-t_0} = \frac{1}{4}(t_1 - t_0)$. See Figure 2.2.

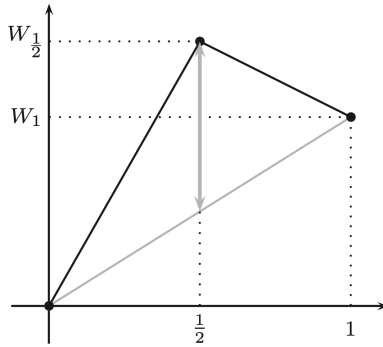


Figure 2.2: Step 2.

3. Continue in the same way as in the step 2 on each interpolate. See Figure 2.3.

At the dyadic points, $t = \frac{k}{2^j}$, we get the true value of $W(t, \omega)$. The piecewise linear functions are successive approximations of the random function $t \mapsto W(t, \omega)$. Let us express the polygon are of Schauder functions. We know that

$$W(t, \omega) = G_0(\omega)S_0(t) + \sum_{j=0}^{\infty} \sum_{k=0}^{2^j-1} G_{2^j+k}(\omega)S_{2^j+k}(t).$$

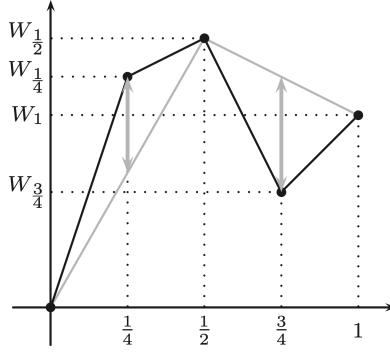


Figure 2.3: Step 3.

For dyadic points $t = \frac{k}{2^j}$, the expansion of W_t is finite. Assume that we already have $W(k/2^j)$ for some fixed $j \geq 1$ and $k = 0, 1, \dots, 2^j$. Then,

$$W(l2^{-j-1}) = \begin{cases} W(k2^{-j}) & \text{if } l = 2k \\ \frac{1}{2}[W(k2^{-j}) + W((k+1)2^{-j}) + \Gamma_{2^j+k}] & \text{if } l = 2k+1 \end{cases}$$

where $\Gamma_{2^j+k} = G_{2^j+k} S_{2^j+k}((2k+1)2^{-j-1})$ is a Gaussian random variable with mean 0 and variance $\frac{1}{4}2^{-j}$. Consequently, each new node is constructed by adding an independent, suitably scaled normal random variable to the linear interpolation of the nodes immediately to the left and right of the new node.

If we calculate $W(1, \omega), W(2, \omega), \dots$, we see that they match with Lévy's construction. Therefore the 2^{th} partial sum $W_{2^j}(t, \omega)$ is the piecewise linear interpolation of the Brownian path $W_t(\omega)$ at the points $(k/2^j, W(k/2^j, \omega))$, $k = 0, \dots, 2^j$.

The following theorem hence says that the sum of the above series is actually a BM¹.

Theorem 2.1.4. *The series*

$$W(t, \omega) := \sum_{n=0}^{\infty} (W_{2^{n+1}}(t, \omega) - W_{2^n}(t, \omega)) + W_1(t, \omega)$$

for $t \in [0, 1]$ converges a.s. uniformly. In particular, $W(t)$ is a BM¹.

In the above theorem, the a.s. convergence ensures the first four distributional properties of being a BM, whereas, the uniform convergence ensures the continuity of the Brownian paths (being the uniform limit of partial sum of continuous functions). Now, the polygonal arc $W_{2^n}(t, \omega)$ can be expressed in terms of Schauder functions as follows:

$$W(t, \omega) = G_0(\omega) S_0(t) + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} G_{2^n+k}(\omega) S_{2^n+k}(t),$$

where,

$$\begin{aligned} G_0(\omega) S_0(t) &= W_1(t, \omega), \\ \sum_{k=0}^{2^n-1} G_{2^n+k}(\omega) S_{2^n+k}(t) &= W_{2^{n+1}}(t, \omega) - W_{2^n}(t, \omega). \end{aligned}$$

and, $S_n(t)$ are the Schauder functions. This shows that Lévy's construction and the Lévy-Ciesielski construction are indeed the same!

2.2 Some basic notions of stochastic processes

Now, we define below some notions about stochastic processes.

Definition 2.2.1. Let $(X_t)_{t \in I}$ and $(Y_t)_{t \in I}$ be two $\mathbb{R}^d$ -valued stochastic processes with the same index set I . We call X and Y

- **indistinguishable** if they are defined on the same probability space and if

$$\mathbb{P}(X_t = Y_t \forall t \in I) = 1,$$

- **modifications** if they are defined on the same probability space and

$$\mathbb{P}(X_t = Y_t) = 1, \quad \forall t \in I,$$

- **equivalent** if they have the same finite dimensional distributions (but X and Y need not be on the same probability space), that is,

$$(X_{t_1}, \dots, X_{t_n}) \sim (Y_{t_1}, \dots, Y_{t_n}),$$

for all $t_1, t_2, \dots, t_n \in I$, $n \geq 1$.

2.3 Wiener's construction

This follows the same idea as Lévy's, where our orthonormal basis of L^2 is instead the trigonometric functions $(e^{in\pi t})_{n \in \mathbb{Z}}$. In this case, the Brownian motion obtained on $[0, 1]$ as

$$\begin{aligned} W(t, \omega) &= \sum_{n=1}^{\infty} \frac{\sin(n\pi t)}{n} G_n(\omega) \\ &= \sum_{n=1}^{\infty} \langle 1_{[0,t]}, \cos(n\pi) \rangle_{L^2} G_n(\omega). \end{aligned}$$

where the latter is the Wiener-Fourier series and $(G_n)_{n \geq 0}$ are the usual standard normal random variables. The convergence results can be analogously proved as we have shown for the Lévy-Ciesielski construction, which will imply that W_t is a Brownian motion.

2.4 Donsker's construction

In this section, we will provide an existential result for Brownian motion, instead of some concrete construction. The main theorem here is the Donsker's invariance principle.

Theorem 2.4.1. Consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and let $(\epsilon_n)_{n \in \mathbb{N}}$ be iid Bernoulli random variables on Ω with $\mathbb{P}(\epsilon_1 = 1) = \frac{1}{2} = \mathbb{P}(\epsilon_1 = -1)$. Define $(S^n(t))_{t \in [0,1]}$ as

$$S^n(t) = \frac{1}{\sqrt{n}} (S_{[nt]} - (nt - [nt])\epsilon_{[nt]+1})$$

where $S_n = \epsilon_1 + \dots + \epsilon_n$. Let $\Phi : \mathcal{C}([0, 1], \mathbb{R}) \rightarrow \mathbb{R}$ be a uniformly continuous bounded linear functional on the linear space $\mathcal{C}[0, 1]$ of $\mathbb{R}$ -valued continuous maps on $[0, 1]$. Then, there exists $(B(t))_{t \in [0,1]}$ a BM¹ such that S^n converges weakly to B (converges in distribution), that is,

$$\lim_{n \rightarrow \infty} \mathbb{E}\Phi(S^n(\cdot)) = \mathbb{E}\Phi(B(\cdot)).$$

Remark 2.4.2. From the statement above, it is clear that a Brownian motion is a limit of linearly interpolated random walks.

Now, let us see why we started off with such a $S^n(t)$.

Proposition 2.4.3. *Consider $(S^n(t))_{t \in [0,1]}$ for $n \in \mathbb{N}$ as defined in the statement of Theorem 2.4.1. Then $(S^n(t))_{t \in [0,1]}$ converges in distribution to a Gaussian process which satisfy first four axioms of a Brownian motion for all rational points in $[0, 1]$.*

Proof. For $j = j(n)$ and $\frac{j}{n} = s = \text{constant}$, the central limit theorem shows that

$$S^n\left(\frac{j}{n}\right) = \frac{\sqrt{s}S_j}{\sqrt{j}} \xrightarrow{d} \sqrt{s}G, \quad (2.3)$$

where G is a standard normal variable. Furthermore, with $s = \frac{j}{n}$, and $t = \frac{k}{n}$, the increment $S^n(t) - S^n(s)$ is independent of $\epsilon_1, \dots, \epsilon_j$, and therefore of all earlier increments. Also, it is easy to calculate

$$\mathbb{E}(S^n(t) - S^n(s)) = 0 \text{ and } \mathbb{V}(S^n(t) - S^n(s)) = \frac{k - j}{n} = t - s.$$

Henceforth, in the limit described in 2.3, we get a Gaussian increment with mean 0 and variance $t - s$. Let us denote the limiting process by $(B_t)_{t \in [0,1]}$, then, as independence and stationarity of the increments are distributional properties, they are inherited by B_t for rational t . So, $(B_q)_{q \in [0,1] \cap \mathbb{Q}}$ satisfy the first four axioms of a BM. $\square$

Now, if B_t had continuous sample paths, then the four axioms would hold for all $t \in [0, 1]$. The fact that $(B_t)_{t \in [0,1]}$ indeed has continuous sample paths follows from the uniform (in t) convergence of $S^n(t)$ given by Donsker's invariance principle. Thus, $(B_t)_{t \in [0,1]}$ is a BM¹.

2.5 The canonical model

In this section we will obtain an analogous model of Brownian motion via natural constructions. The main idea is that we want to identify the probability space $(\Omega, \mathcal{A}, \mathbb{P})$ on which our Brownian motion is defined to a subspace of $(\mathbb{R}^D)^I$ so that we can get rid of the ambient space Ω , which highly simplifies our life as we then have to just work with the sample paths defined by a Brownian motion. So, let us start the construction.

Any $\mathbb{R}^d$ -valued stochastic process $(X(t))_{t \in I}$ is a map

$$\begin{aligned} I \times \Omega &\longrightarrow \mathbb{R}^d \\ (t, \omega) &\longmapsto X(t, \omega). \end{aligned}$$

We can also view this map as

$$\begin{aligned} \Omega &\longrightarrow (\mathbb{R}^d)^I = \{w : w : I \rightarrow \mathbb{R}^d\} \\ \omega &\longmapsto \{t \mapsto X(t, \omega)\}. \end{aligned}$$

If we identify Ω with a subset of $(\mathbb{R}^d)^I$, then, we get the so-called *canonical model*. For Brownian motion, we consider the subspace $\mathcal{C}_{(0)}[0, \infty) \subset (\mathbb{R}^d)^{[0, \infty)}$ (as we want to ensure the continuity of the paths), where,

$$\mathcal{C}_{(0)} := \mathcal{C}_{(0)}[0, \infty) := \{w : I \rightarrow \mathbb{R}^d \text{ is continuous and } w(0) = 0\}.$$

Now, we want to equip this space with an appropriate measure and a sigma algebra. For that, we consider the following maps (projection onto the t -th coordinate): For $I = [0, \infty)$,

$$\begin{aligned}\pi_t : (\mathbb{R}^d)^I &\rightarrow \mathbb{R}^d \\ w &\mapsto w(t).\end{aligned}$$

Now, to see what appropriate sigma algebra we can equip this space with, let us first consider the canonical sigma algebra of the space $(\mathbb{R}^d)^I$:

$$\mathcal{B}^I(\mathbb{R}^d) = \sigma\{\pi_t^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^d), t \in I\} = \sigma\{\pi_t : t \in [0, 1]\}.$$

One might now be tempted to consider the restriction of this sigma algebra to the space $\mathcal{C}_I$, but, unfortunately, the space $\mathcal{C}_{(0)}$ doesn't lie inside $\mathcal{B}^I(\mathbb{R}^d)$. So, we need to consider the trace σ -algebra

$$\mathcal{C}_{(0)} \cap \mathcal{B}^I(\mathbb{R}^d) = \sigma\{\pi_t|_{\mathcal{C}_{(0)}} : t \in [0, 1]\}.$$

Now, note that if we consider the metric of local uniform convergence,

$$\rho(w, v) = \sum_{n=1}^{\infty} \left(1 \wedge \sup_{0 \leq t \leq n} |w(t) - v(t)| \right) 2^{-n},$$

then, $\mathcal{C}_{(0)}$ equipped with this metric, becomes a complete and separable metric space.

Lemma 2.5.1. *If $\mathcal{O}_\rho$ is the topology induced by ρ , then, it can be proved that the Borel σ -algebra $\mathcal{B}(\mathcal{C}_{(0)}) := \sigma(\mathcal{O}_\rho)$ is equal to the previously defined trace σ -algebra, that is,*

$$\mathcal{C}_{(0)} \cap \mathcal{B}^I(\mathbb{R}^d) = \mathcal{B}(\mathcal{C}_{(0)}).$$

$\mathcal{B}(\mathcal{C}_{(0)})$ is the sigma algebra that we equip our space $\mathcal{C}_{(0)}$ with. Now, let us figure out the measure that we want to equip our space $\mathcal{C}_{(0)}$ with. For any BM^d $(B_t)_{t \geq 0}$, we consider the map

$$\begin{aligned}\psi : \Omega &\rightarrow \mathcal{C}_{(0)} \\ \omega &\mapsto w := \psi(\omega) = B(-, \omega).\end{aligned}$$

Note that

1. The metric balls $\mathbb{B}^\rho(v, r) \subset \mathcal{C}_{(0)}$ generate $\mathcal{B}(\mathcal{C}_{(0)})$, and
2. $\{\omega : \psi(\omega) \in \mathbb{B}^\rho(v, r)\} = \{\omega : \rho(B(-, \omega), v) < r\}$,

whence, the map

$$\begin{aligned}\omega &\mapsto \rho(B(-, \omega), w) \\ &= \sum_{n=1}^{\infty} \{1 \wedge \sup_{t \in [0, n]} |B(t, \omega) - w(t)|\} 2^{-n} \\ &= \sum_{n=1}^{\infty} \{1 \wedge \sup_{t \in [0, n]} |\pi_t(B(-, \omega)) - \pi_t(w)|\} 2^{-n} \quad \text{is } \mathcal{A} \text{ measurable,}\end{aligned}$$

since, $\mathcal{B}(\mathcal{C}_{(0)})$ is the smallest sigma algebra wrt which $\pi_t(w)$ is measurable for any $w \in \mathcal{C}_{(0)}$. Therefore, it follows that ψ is $\mathcal{A}/\mathcal{B}(\mathcal{C}_{(0)})$ measurable, and, the pushforward measure

$$\mu(\Gamma) := \mathbb{P}(\psi^{-1}(\Gamma)), \quad \Gamma \in \mathcal{B}(\mathcal{C}_{(0)}),$$

is a measure on $\mathcal{C}_{(0)}$. Now, since the cylinder sets

$$\Gamma = \{w \in \mathcal{C}_{(0)} : \pi_{t_1}(w) \in C_1, \dots, \pi_{t_n}(w) \in C_n\},$$

are $\cap$ -stable generators of $\mathcal{B}(\mathcal{C}_{(0)})$, for all $n \geq 1$, $0 < t_1 < \dots < t_n$, and $C_1, \dots, C_n \in \mathcal{B}(\mathbb{R}^d)$, it follows that the finite dimensional distributions

$$\mu(\Gamma) = \mu(\pi_{t_1} \in C_1, \dots, \pi_{t_n} \in C_n) = \mathbb{P}(B(t_1) \in C_1, \dots, B(t_n) \in C_n), \quad (2.4)$$

uniquely determine the measure μ . Henceforth, we have the following theorem:

Theorem 2.5.2. *On the probability space $(\mathcal{C}_{(0)}, \mathcal{B}(\mathcal{C}_{(0)}), \mu)$, the family of projections $(\pi_t)_{t \geq 0}$ satisfies all the axioms in the definition of BM^d , and thus, $(\pi_t)_{t \geq 0}$ is a Brownian motion.*

The first four distributional properties are clear from equation 2.4, and the continuity of the paths follow as we are considering projections (continuous functions) π_t from the space the space $\mathcal{C}_{(0)}$. So the upshot here is that every Brownian motion $(\Omega, \mathcal{A}, \mathbb{P}, B_t, t \geq 0)$ admits an equivalent Brownian motion $(W_t)_{t \geq 0}$ on the probability space $(\mathcal{C}_{(0)}, \mathcal{B}(\mathcal{C}_{(0)}), \mu)$.

Definition 2.5.3. The space $(\mathcal{C}_{(0)}, \mathcal{B}(\mathcal{C}_{(0)}), \mu, \pi_t)$ is called the *Wiener space* or *path space*, the measure μ is called the *Wiener measure*, and $(\pi_t)_{t \geq 0}$ is called the the canonical (model of the) Brownian motion process.

2.6 Kolmogorov's Construction

This method of construction relies on using *Kolmogorov's consistency theorem* (and some further results) for obtaining a Brownian motion. Let us see how to do it.

As we saw above, the measure μ is uniquely determined by the finite dimensional projections

$$\pi_{t_1, \dots, t_n}(w) := (w(t_1), \dots, w(t_n))$$

and the corresponding finite dimensional distributions

$$p_{t_1, \dots, t_n}(C_1 \times \dots \times C_n) = \mu(\pi_{t_1, \dots, t_n} \in C_1 \times \dots \times C_n),$$

where $t_1, \dots, t_n \in I$ and $C_1, \dots, C_n \in \mathcal{B}(\mathbb{R}^d)$. Now, if $p_{t_1, \dots, t_n}$, $n \geq 1$, $t_1, \dots, t_n \in I$ are finite dimensional distributions of a stochastic process, then, it should necessarily satisfy the following two conditions (as the family of finite dimensional distributions of any stochastic process satisfy these conditions):

1. $p_{t_1, \dots, t_n}(C_1 \times \dots \times C_n) = p_{t_{\sigma(1)}, \dots, t_{\sigma(n)}}(C_{\sigma(1)} \times \dots \times C_{\sigma(n)})$, for all permutations $\sigma : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$.
2. $p_{t_1, \dots, t_{n-1}, t_n}(C_1 \times \dots \times C_{n-1} \times \mathbb{R}^d) = p_{t_1, \dots, t_{n-1}}(C_1 \times \dots \times C_{n-1})$,

for all $n \geq 1$, $t_1, \dots, t_n \in I$.

Note: The above conditions are called the *consistency conditions*.

Definition 2.6.1. For all $n \geq 1$ and all $t_1, \dots, t_n \in I$, if the family of probability measures $p_{t_1, \dots, t_n}$ satisfy the consistency conditions, then, we call them *consistent* or *projective*.

Theorem 2.6.2 (Kolmogorov's consistency theorem, 1933). *Let $I \subset [0, \infty)$ and $p_{t_1, \dots, t_n}$ be probability measures defined on $((\mathbb{R}^d)^n, \mathcal{B}((\mathbb{R}^d)^n))$ for all $t_1, \dots, t_n \in I$, $n \geq 1$. If the family of measures is consistent, then there exists a probability measure μ on $((\mathbb{R}^d)^n, \mathcal{B}((\mathbb{R}^d)^n))$ such that*

$$p_{t_1, \dots, t_n}(C) = \mu(\pi_{t_1, \dots, t_n}^{-1}(C)), \text{ for all } C \in \mathcal{B}((\mathbb{R}^d)^n).$$

Now, to construct the Brownian motion, for $d = 1$, and $0 < t_1 < \dots < t_n$, we consider the n -variate Gaussian measures

$$p_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sqrt{\det C}} \int_{A_1 \times \dots \times A_n} e^{-\frac{1}{2}\langle x, C^{-1}x \rangle} dx,$$

with $C = (t_j \wedge t_k)_{j,k=1, \dots, n}$. It is an easy check that the above family of distribution functions satisfy the consistency conditions. This proves that there exists a $\mathbb{R}$ -valued stochastic process $(B_t)_{t \geq 0}$ which satisfy the first four axioms in the definition of a Brownian motion. Now, for ensuring the continuity of the sample paths, we use another theorem by Kolmogorov:

Theorem 2.6.3 (Kolmogorov 1934; Slutsky 1937; Chentsov 1956). *Consider a stochastic process $(X_t)_{t \geq 0}$ a stochastic process on $(\Omega, \mathcal{A}, \mathbb{P})$ taking values in $\mathbb{R}^d$. If*

$$\mathbb{E}(|X(t) - X(s)|^\alpha) \leq c|t - s|^{1+\beta} \text{ for all } s, t \geq 0$$

holds for some constant $c > 0$ and $\alpha, \beta > 0$, then, $(X_t)_{t \geq 0}$ has a modification $(X'_t)_{t \geq 0}$ which has only continuous sample paths.

Note that for the process $(B_t)_{t \geq 0}$ satisfying the first four axioms, we can take $\alpha = 4$, and $\beta = 1$,

$$\mathbb{E}(|B(t) - B(s)|^4) = 3|t - s|^2,$$

and thus, $(B_t)_{t \geq 0}$ is a Brownian motion.

Chapter 3

Brownian motion as a martingale

We have seen before that Brownian motion have Markov property. In this chapter we will see how we can view Brownian motion as a martingale. Before we start elaborating on this, let us first recall the definitions of filtration and martingale.

Let I be some subset of $[0, \infty]$ and $(\Omega, \mathcal{A}, \mathbb{P})$ be our probability space.

Definition 3.0.1 (Filtration). A family $(\mathcal{F}_t)_{t \in I}$ is called a filtration if it is an increasing family of sub σ -algebras of $\mathcal{A}$.

Definition 3.0.2 (Martingale). A martingale $(X_t, \mathcal{F}_t)_{t \in I}$ is a real or complex stochastic process $X_t : \Omega \rightarrow \mathbb{R}^d$ or $X_t : \Omega \rightarrow \mathbb{C}$ satisfying

1. $\mathbb{E}|X_t| < \infty$ for all $t \in I$;
2. X_t is $\mathcal{F}_t$ measurable for all $t \in I$;
3. $\mathbb{E}(X_t | \mathcal{F}_s) = X_s$ for all $s, t \in I, s \leq t$.

Definition 3.0.3 (Sub-or supermartingale). If X_t is real valued, and in property (3) of the definition of a martingale if we have ' $\geq$ ' or ' $\leq$ ', then $(X_t, \mathcal{F}_t)_{t \in I}$ is called sub-or supermartingale respectively.

Definition 3.0.4 (Adapted stochastic process). A stochastic process $(X_t)_{t \in I}$ is said to be adapted to the filtration $(\mathcal{F}_t)_{t \in I}$ if for each $t \in I$, X_t is $\mathcal{F}_t$ -measurable.

Clearly, a stochastic process $(X_t)_{t \in I}$ is $(\mathcal{F}_t)_{t \in I}$ adapted if for each $t \in I$, $\mathcal{F}_t^X := \sigma(X_s : s \leq t) \subset \mathcal{F}_t$. Here, $\mathcal{F}_t^X$ is called the natural filtration of X .

Since, from the Markov property of the Brownian motion B_t , we have

$$B_t - B_s \perp \mathcal{F}_s^B \text{ for all } 0 \leq s \leq t, \quad (3.1)$$

it follows that a Brownian motion is a martingale with respect to its natural filtration $\mathcal{F}_t^B$. The first two properties of the definition are clear; for property (3),

$$\mathbb{E}(B_t | \mathcal{F}_s) = \mathbb{E}(B_t - B_s | \mathcal{F}_s) + \mathbb{E}(B_s | \mathcal{F}_s) = \mathbb{E}(B_t - B_s) + B_s = B_s,$$

where we have used equation (3.1), and the fact that on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$, if X is a random variable which is independent of a sub σ -algebra $\mathcal{G}$ of $\mathcal{A}$, then, $\mathbb{E}(X | \mathcal{G}) = E(X)$ and if Y is a $\mathcal{G}$ measurable random variable, then, $E(Y | \mathcal{G}) = Y$.

Now, it is often necessary to enlarge the canonical filtration $(\mathcal{F}_t^B)_{t \in I}$ of a Brownian motion $(B_t)_{t \in I}$, for which let us define something called a admissible filtration.

Definition 3.0.5 (Admissible filtration). Let $(B_t)_{t \geq 0}$ be a d -dimensional Brownian motion. A filtration $(\mathcal{F}_t)_{t \geq 0}$ is called admissible, if

1. $\mathcal{F}_t^B \subset \mathcal{F}_t$ for all $t \geq 0$,
2. $B_t - B_s \perp \mathcal{F}_s$ for all $0 \leq s \leq t$.

Also, if $\mathcal{F}_0$ contains all the $\mathbb{P}$ null sets, then, $(\mathcal{F}_t)_{t \geq 0}$ is called admissible complete filtration.

It is clear that the canonical filtration corresponding to $(B_t)_{t \geq 0}$ is always admissible. The main point of considering something like a admissible filtration is that the canonical filtration of $(B_t)_{t \geq 0}$ doesn't provide us with much interesting results, but with respect to an admissible filtration of $(B_t)_{t \geq 0}$, we can construct varieties of 'nice' martingales out of the process $(B_t)_{t \geq 0}$.

Example 3.0.6. Let $(B_t)_{t \geq 0}$ be a d -dimensional Brownian motion and $(\mathcal{F}_t)_{t \geq 0}$ be an admissible filtration.

1. $(B_t)_{t \geq 0}$ is a martingale with respect to $\mathcal{F}_t$.
2. $M_t := |B_t|^2$ is a positive submartingale with respect to $\mathcal{F}_t$. This easily follows by using the conditional form of Jensen's inequality, which states: If ϕ is a convex function, then,

$$\mathbb{E}[\phi(X)|\mathcal{F}] \geq \phi(\mathbb{E}[X|\mathcal{F}]),$$

where $\mathcal{F}$ is a sub σ -algebra of $\mathcal{A}$.

3. $M_t := |B_t|^2 - d \cdot t$ is a martingale with respect to $\mathcal{F}_t$. Since, $M_t = \sum_{j=1}^d [(B_t^j)^2 - t]$, it suffices to consider $d = 1$. Then, using the properties of conditional expectation, and the fact that $B_t - B_s \perp \mathcal{F}_s$ for all $0 \leq s \leq t$, the required follows.
4. $M_t^\xi := e^{i\langle \xi, B(t) \rangle + \frac{t}{2}|\xi|^2}$, for all $\xi \in \mathbb{R}^d$ is a complex valued martingale with respect to $\mathcal{F}_t$.

Let us state a lemma which will be useful in proving some major results ahead.

Lemma 3.0.7. Let $(B_t)_{t \geq 0}$ be a d -dimensional Brownian motion. Then we have for $t \geq 0$ and $p > 1$

$$\mathbb{E} \left[\sup_{s \leq t} |B_s|^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|B_t|^p].$$

Now, as we have gathered some knowledge about martingales and filtrations, let us give a characterization of Brownian motion using conditional characteristic functions.

Lemma 3.0.8 (Characterization of Brownian motion through conditional characteristic functions). Let $(X_t)_{t \geq 0}$ be a $\mathbb{R}^d$ -valued continuous stochastic process with $X_0 = 0$ and which is adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$. Then $(X_t, \mathcal{F}_t)_{t \geq 0}$ is a Brownian motion, if and only if, for all $0 \leq s \leq t$ and $\xi \in \mathbb{R}^d$

$$\mathbb{E}[e^{i\langle \xi, X_t - X_s \rangle} | \mathcal{F}_s] = e^{-\frac{1}{2}|\xi|^2(t-s)}.$$

Proof. To proof this, we use the characterization of BM that we mentioned earlier in chapter 1, that is, a stochastic process $(X_t)_{t \geq 0}$ with continuous sample paths is said to be a BM if for all $n \geq 0$, $0 = t_0 < t_1 < \dots < t_n$ and $\xi_0, \xi_1, \dots, \xi_n \in \mathbb{R}^d$, and $X_{t_{-1}} := 0$,

$$\mathbb{E} \left[\exp \left(i \sum_{j=0}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle \right) \right] = \exp \left[-\frac{1}{2} \sum_{j=1}^n |\xi_j|^2 (t_j - t_{j-1}) \right].$$

Now, using the tower property and the fact that the random variable $X_{t_n} - X_{t_{n-1}}$ is independent of the sigma algebra $\mathcal{F}_{t_{n-1}}$ for the only if part and the hypothesis for the if part, we have

$$\begin{aligned} \mathbb{E} \left[\exp \left(i \sum_{j=0}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle \right) \right] &= \mathbb{E} \left[\mathbb{E} \left[\exp \left(i \langle \xi_n, X_{t_n} - X_{t_{n-1}} \rangle \right) \mid \mathcal{F}_{t_{n-1}} \right] \exp \left(i \sum_{j=0}^{n-1} \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle \right) \right] \\ &= \exp \left(-\frac{1}{2} |\xi_n|^2 (t - s) \right) \mathbb{E} \left[\exp \left(i \sum_{j=0}^{n-1} \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle \right) \right]. \end{aligned}$$

Iterating this step, we obtain our desired result. $\square$

3.1 Stopping and sampling

We have previously encountered the concept of stopping time in our introductory martingale lectures. Now, to generalize it in the measure-theoretic sense, let us first recall the importance of considering these stopping times. To analyze the time points at which our Brownian motion shows certain interesting behaviour such as when a Brownian motion $(B_t)_{t \geq 0}$ leaves or enters a particular set for the first time, hits its running maximum or say, returns to zero, we have to look at random times. A stopping time $\tau : \Omega \rightarrow [0, \infty)$ is a random time exhibiting the property

$$\{\omega \in \Omega : \tau(\omega) \leq t\} \in \mathcal{F}_t \text{ for all } t \geq 0,$$

which basically says that the event $\{\tau \leq t\}$, is completely determined by the random variables $X_s, s \leq t$.

Some examples of stopping times are entry and hitting times of a process $(X_t)_{t \geq 0}$ into a set $A \in \mathcal{B}(\mathbb{R}^d)$ defined as:

1. first entry time into A: $\tau_A^\circ(\omega) := \inf\{t \geq 0 : X_t(\omega) \in A\}$,
2. first hitting time of A: $\tau_A(\omega) := \inf\{t > 0 : X_t(\omega) \in A\}$,

where if the sets are empty, then we define $\inf \phi := \infty$. It is clear that the first exit time from A will be then analogously defined as τ_{A^c} . Also it is an easy observation that $\tau_A^\circ \leq \tau_A$. It is only when our sample paths $t \rightarrow X_t$ and filtration $(\mathcal{F}_t)_{t \geq 0}$ in consideration are sufficiently regular that the above random times are stopping times for every Borel set $\mathbb{R}^d$, so in our discussion, we will be restricted to open and closed sets A.

For many interesting stopping times, just considering the natural filtration of our given stochastic process won't work, as it's relatively small. So, we consider a slightly larger filtration

$$\mathcal{F}_{t+}^X := \bigcap_{u>t} \mathcal{F}_u^X = \bigcap_{n \geq 1} \mathcal{F}_{t+\frac{1}{n}}^X.$$

For example, we show below that the entry time and the hitting time defined above are both $\mathcal{F}_{t+}^X$ stopping times.

Lemma 3.1.1. *Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process with right continuous sample paths and $U \subset \mathbb{R}^d$ an open set. The first hitting time τ_U satisfies*

$$\{\omega : \tau_U(\omega) < t\} \in \mathcal{F}_t^X \text{ and } \{\omega : \tau_U(\omega) \leq t\} \in \mathcal{F}_{t+}^X.$$

Proof. To prove the first assertion, let us first prove the following equality:

$$\{\omega \in \Omega : \tau_U(\omega) = \inf\{s > 0 : X_s(\omega) \in U\} < t\} = \bigcup_{\mathbb{Q}^+ \ni r \leq t} \{\omega \in \Omega : X_r(\omega) \in U\}. \quad (3.2)$$

For the ' $\subset$ ' part: For any $\omega \in \Omega$ if $\tau_U(\omega) < t$, there exists some $s < t$ such that $X_s(\omega) \in U$. Since the paths are right continuous and U is an open set, we can always find some $r \in \mathbb{Q}^+$ such that $s < r < t$ and $X_r(\omega) \in U$.

For the ' $\supset$ ' part: Suppose for any $\omega \in \Omega$, $X_r(\omega) \in U$, for some $r \in (0, t]$. Since, U is open and $t \rightarrow X_t$ is right continuous, we can always find a small enough neighbourhood around r such that for all the points, say q in that neighbourhood $X_q(\omega)$ will lie in U . Since r is arbitrary here, we will always have $\tau_U(\omega) < r < t$.

Now, clearly in the RHS of equation (3.2), each set in the union belongs to $\mathcal{F}_r^X$, and hence, the union is in $\mathcal{F}_t^X$. So, we have established that $\{\omega : \tau_U(\omega) < t\} \in \mathcal{F}_t^X$.

The second assertion also follows easily from the first, since $\{\tau_U \leq t\} = \bigcap_{n \geq 1} \{\tau_U < t + \frac{1}{n}\} \in \bigcap_{n \geq 1} \mathcal{F}_{t+\frac{1}{n}}^X = \mathcal{F}_{t+}^X$. $\square$

Remark 3.1.2. Using the properties of open sets, it is easy to see that τ_U° and τ_U coincide, hence, τ_U° is also a $\mathcal{F}_{t+}^X$ stopping time.

Lemma 3.1.3. Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process with continuous sample paths and $F \subset \mathbb{R}^d$ be a closed set. Then τ_F° is a $\mathcal{F}_t^X$ stopping time whereas τ_F is a $\mathcal{F}_{t+}^X$ stopping time.

Proof. We make use of the uniformly continuous function $d(x, F)$, which measures the distance of x from the closed set F . We show that the following equality holds:

$$\{\omega \in \Omega : \tau_F^\circ(\omega) \leq t\} = \left\{ \omega \in \Omega : \inf_{r \in \mathbb{Q} \cap [0, t]} d(X_r(\omega), F) = 0 \right\} =: \Omega_t. \quad (3.3)$$

Showing the above equality suffices to show the first assertion of the hypothesis, since for each r , the event $\{\omega \in \Omega : d(X_r(\omega), F) = 0\}$ is a $\mathcal{F}_r$ measurable event, and hence, Ω_t is $\mathcal{F}_t$ measurable. For the backward direction of this equality, consider $\omega \in \Omega$ such that $\inf_{r \in \mathbb{Q} \cap [0, t]} d(X_r(\omega), F) = 0$, then we can find a sequence $(r_j)_{j \geq 1}$ such that $r_j \rightarrow s$ as $j \rightarrow \infty$, $d(X_{r_j}(\omega), F) \rightarrow 0$ as $j \rightarrow \infty$ and since the sample paths are continuous, $X(r_j, \omega) \rightarrow X(s, \omega)$ as $j \rightarrow \infty$. Using the continuity of the distance function, we have $d(X_s, F) = 0$, and since F is a closed set, $\tau_F^\circ(\omega) \leq s \leq t$. The forward direction of this equality follows easily by going through contradiction.

Now, let us prove that the sets $\{\tau_F \leq t\}$ are $\mathcal{F}_{t+}^X$ measurable. As we have seen before, if we can prove that the sets $\{\tau_F < t\}$ are $\mathcal{F}_t$ measurable, then we are done. So, we define the following sets:

$$\tau_n := \inf\{s \geq 0 : X_{s+\frac{1}{n}} \in F\}.$$

Then,

$$\tau_F = \inf_{n \geq 1} \inf\{t \geq \frac{1}{n} : X_t \in F\} = \inf_{n \geq 1} \left(\tau_n + \frac{1}{n} \right),$$

and,

$$\{\tau_F < t\} = \bigcup_{n \geq 1} \left\{ \tau_n \leq t - \frac{1}{n} \right\} \in \mathcal{F}_t^X.$$

$\square$

3.1.1 Point recurrence property of Brownian motion

Let us first define the term *first passage time* of a BM. Let $(B_t)_{t \geq 0}$ be a BM¹, then the first entry time $\tau_b := \tau_{\{b\}}^\circ$ into the closed set $\{b\}$ is known as the first passage time.

Now, we have

$$\sup_{t \geq 0} B_t \geq \sup_{n \geq 1} B_n = \sup_{n \geq 1} (\xi_1 + \cdots + \xi_n) =: S,$$

where $\xi_j := B_j - B_{j-1}$ are iid standard normal random variables, and due to which $S = \infty$ a.s.

The same argument when applied to $\inf_{t \geq 0} B_t$, we obtain

$$\mathbb{P} \left(\sup_{t \geq 0} B_t = +\infty, \inf_{t \geq 0} B_t = -\infty \right) = 1, \quad (3.4)$$

and hence, $\mathbb{P}(\tau_b < \infty) = 1$.

Also, because of the continuity of the sample paths, from equation 3.4, which tells that the sample paths of the Brownian motion always oscillates between $+\infty$ and $-\infty$, it directly follows that the random set $\{t \geq 0 : B_t(\omega) = b\}$ is a.s. unbounded, for all $b \in \mathbb{R}$, as the sample paths takes every real value infinitely many times. In particular, a Brownian motion is point recurrent.

3.2 Wald's Identities

To look at the Wald's identities, let us recall the optional stopping time theorem.

3.2.1 Optional stopping time theorem

Before we state and prove the next famous theorem, let us first see the optional stopping time theorem, here in the case of continuous time martingales.

Theorem 3.2.1. *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a martingale with continuous sample paths and let $\sigma \leq \tau$ be $\mathcal{F}_t$ stopping times. Then, we have that for all $k \geq 0$, $X_{\sigma \wedge k}, X_{\tau \wedge k} \in L^1(\mathbb{P})$, and*

$$\mathbb{E}(X_{\tau \wedge k} | \mathcal{F}_{\sigma \wedge k}) = X_{\sigma \wedge k}.$$

Moreover, if $\sigma \leq \tau$ are bounded $\mathcal{F}_t$ stopping times, that is, τ is an a.s. bounded stopping time, $X_\sigma, X_\tau \in L^1(\mathbb{P})$, and $\lim_{k \rightarrow \infty} \mathbb{E}[|X_k(\omega)| \chi_{\{\omega: \tau(\omega) > k\}}] = 0$, then

$$\mathbb{E}(X_\tau | \mathcal{F}_\sigma) = X_\sigma.$$

3.2.2 Wald's identity

Theorem 3.2.2. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM¹ and assume that τ is an $\mathcal{F}_t$ stopping time. If $\mathbb{E}\tau < \infty$, then $B_\tau \in L^2(\mathbb{P})$ and we have*

$$\mathbb{E} B_\tau = 0, \quad \text{and} \quad \mathbb{E} B_\tau^2 = \mathbb{E} \tau.$$

Proof. We will prove this theorem in the following four steps:

1. $(B_{\tau \wedge t}, F_{\tau \wedge t})_{t \geq 0}$ is a martingale.
2. $B_{\tau \wedge t} \in L^2(\mathbb{P})$.
3. $B_{\tau \wedge t}$ is Cauchy in $L^2(\mathbb{P})$.

4. Obtain $\mathbb{E} B_\tau$ and $\mathbb{E} B_\tau^2$ by realising them as a limit of $\mathbb{E} B_{\tau \wedge t}$ and $\mathbb{E} B_{\tau \wedge t}^2$ respectively.

Part (1) follows immediately from optional stopping theorem stated above as $(t \wedge \tau)_{t \geq 0}$ is a family of bounded stopping times. Also, this implies $\mathbb{E} B_{\tau \wedge t} = \mathbb{E} B_0 = 0$.

Part (2): From 3.0.7,

$$\mathbb{E} B_{\tau \wedge t}^2 \leq \mathbb{E} [\sup_{s \leq t} B_{\tau \wedge s}^2] \leq 4 \mathbb{E} [B_t^2] = 4t.$$

So, now we can talk about $\mathbb{E} B_{\tau \wedge t}^2$, and from part (3) of example 3.0.6, it follows that $(M_t := B_{\tau \wedge t}^2 - \tau \wedge t)_{t \geq 0}$ is a martingale, and hence,

$$\mathbb{E} B_{\tau \wedge t}^2 = \mathbb{E} \tau \wedge t.$$

Part (3): Note that by property of martingale and Tower property of conditional expectations, $\mathbb{E} [B_{\tau \wedge t} B_{\tau \wedge s}] = \mathbb{E} [B_{\tau \wedge s}^2]$, for all $s \leq t$, and hence as $\mathbb{E} \tau < \infty$, by DCT, it follows that

$$\mathbb{E} [(B_{\tau \wedge t} - B_{\tau \wedge s})^2] = \mathbb{E} [\tau \wedge t - \tau \wedge s] \longrightarrow 0,$$

as $s, t \rightarrow \infty$. So, $(B_{\tau \wedge t})_{t \geq 0}$ converges in L^2 , whence by the continuity of Brownian sample paths and the fact that $\tau < \infty$ a.s., $L^2 - \lim_{t \rightarrow \infty} B_{\tau \wedge t} = B_\tau$. In particular, B_τ is in $L^2(\mathcal{P})$ and thus, by monotone convergence theorem,

$$\mathbb{E} B_\tau^2 = \lim_{t \rightarrow \infty} \mathbb{E} [B_{\tau \wedge t}^2] = \lim_{t \rightarrow \infty} \mathbb{E} [\tau \wedge t] = \mathbb{E} \tau.$$

Also, since, L^2 convergence implies L^1 convergence, in the same way it follows that $\mathbb{E} B_\tau = 0$. $\square$

Corollary 3.2.3. *Let $(B_t)_{t \geq 0}$ be a BM^1 and $\tau = \inf\{t \geq 0 : B_t \notin (a, b)\}$ be the first entry time into the set $(-a, b)^c$, $a, b \geq 0$. Then*

$$\mathbb{P}(B_\tau = -a) = \frac{b}{a+b}, \quad \mathbb{P}(B_\tau = b) = \frac{a}{a+b}, \quad \text{and} \quad \mathbb{E} \tau = ab.$$

Now let us look at a very strange corollary to theorem 3.2.2 which says that a Brownian motion hits 1 infinitely often but the average time the BM needs to get there is infinite.

Corollary 3.2.4. *Let $(B_t)_{t \geq 0}$ be a BM^1 and τ_1 be the first passage time of the level 1. Then $\mathbb{E} \tau = \infty$.*

Proof. We have already seen that the first passage time $\tau_1 < \infty$ a.s. Then, $B_{\tau_1} = 1$ which implies the expectation $\mathbb{E} B_{\tau_1} = 1$ and the second moment $\mathbb{E} B_{\tau_1}^2 = 1$. But by Wald's identity, this can only happen if $\mathbb{E} \tau = \infty$. $\square$

Now let us look at another Wald's identity.

3.2.3 Exponential Wald Identity

Theorem 3.2.5. *Let $(B_t)_{t \geq 0}$ be a BM^1 . Then the first passage time τ_b , for any $b \in \mathbb{R}$ is a.s. finite and its Laplace transform is given by*

$$\mathbb{E} e^{-\zeta \tau_b} = e^{-\sqrt{2\zeta}|b|},$$

for all $\zeta \geq 0$.

Proof. As we have seen before, $M^\xi(t) := e^{\xi B(t) - \frac{1}{2}\xi^2 t}$ is martingale, and also $(t \wedge \tau_b)_{t \geq 0}$ is a family of bounded stopping times. Hence, by optional stopping time theorem,

$$\mathbb{E} [e^{\xi B(t \wedge \tau_b) - \frac{1}{2}\xi^2 (t \wedge \tau_b)}] = \mathbb{E} M^\xi(0) = 1. \quad (3.5)$$

Now, observing

$$e^{\xi B(t \wedge \tau_b) - \frac{1}{2}\xi^2 (t \wedge \tau_b)} \leq e^{\xi b},$$

and that

$$\lim_{t \rightarrow \infty} e^{\xi B(t \wedge \tau_b) - \frac{1}{2} \xi^2 (t \wedge \tau_b)} = \begin{cases} e^{\xi B(\tau_b) - \frac{1}{2} \xi^2 (\tau_b)} & \text{if } \tau_b < \infty \\ 0 & \text{if } \tau_b = \infty, \end{cases}$$

from DCT it follows that, as $t \rightarrow \infty$

$$1 = \mathbb{E} [e^{\xi B(t \wedge \tau_b) - \frac{1}{2} \xi^2 (t \wedge \tau_b)}] \rightarrow \mathbb{E} [\chi_{\{\tau_b < \infty\}} e^{\xi B(\tau_b) - \frac{1}{2} \xi^2 (\tau_b)}],$$

that is,

$$\mathbb{E} [\chi_{\{\tau_b < \infty\}} e^{-\frac{1}{2} \xi^2 (\tau_b)}] = e^{-\xi b}.$$

Taking $\xi \rightarrow 0$ both sides of this equation and using monotone convergence theorem, as $e^{-\frac{1}{2} \xi^2 \tau_b}$ is a decreasing function in ξ , we obtain $\mathbb{P}(\tau_b < \infty)$. Using this in the above equality, we obtain our result for $b \geq 0$ and for $b < 0$, the proof follows analogously by replacing ξ with $-\xi$. $\square$

Theorem 3.2.6 (Exponential Wald identity). *Let $(B_t)_{t \geq 0}$ be a BM^1 and let τ be $\mathcal{F}_t^B$ stopping time such that $\mathbb{E} e^{\tau/2} < \infty$. Then*

$$\mathbb{E} e^{B_\tau - \frac{1}{2} \tau} = 1.$$

Proof. Note that we already have

$$\mathbb{E} e^{B_\tau - \frac{1}{2} \tau} \leq \mathbb{E} \left[\liminf_{t \rightarrow \infty} e^{B_{\tau \wedge t} - \frac{1}{2} (\tau \wedge t)} \right] \leq \liminf_{t \rightarrow \infty} \mathbb{E} \left[e^{B_{\tau \wedge t} - \frac{1}{2} (\tau \wedge t)} \right] = 1. \quad (3.6)$$

If we can show that $\mathbb{E} e^{B_\tau - \frac{1}{2} \tau}$ is also lower bounded by 1, then we are done. For that, we consider any $c \in (0, 1)$ and fix it and $p = p(c)$ such that $p < \frac{1}{c} \wedge \frac{1}{c(2-c)}$. Also, consider the martingale $M_t := e^{cB_{\tau \wedge t} - \frac{1}{2} c^2 (\tau \wedge t)}$. We show that $\{M_t^p\}$ is uniformly integrable for all p within the above mentioned bounds. In the course of the proof, we will be able to see its relevance.

For the conjugate exponents $\frac{1}{pc}$ and $\frac{1}{1-pc}$, we have by Hölder's inequality,

$$\mathbb{E} \left[e^{pcB_{\tau \wedge t} - \frac{1}{2} pc^2 (\tau \wedge t)} \right] \leq \left[\mathbb{E} e^{B_{\tau \wedge t} - \frac{1}{2} \tau \wedge t} \right]^{pc} \left[\mathbb{E} e^{\frac{1}{2} \frac{pc(1-pc)}{1-pc} \tau} \right]^{1-pc} \leq (\mathbb{E} e^{\tau/2})^{1-pc},$$

because of the martingale property of M_t and the fact that $\frac{pc(1-c)}{1-pc} \leq 1$ for all p in consideration. This shows that the p -th moment of the martingale M_t is uniformly bounded for all $t > 0$, and hence, it follows that $\{M_t^p\}$ is uniformly integrable. Now, from the result in probability theory that states if for any $r \geq 1$, we have $\{|X_t|^r\}$ is uniformly integrable, then $\mathbb{E}(X_t) \rightarrow \mathbb{E}(X)$, for any collection of random variables $(X_t)_{t \geq 0}$ such that X_t converges to X a.s. Since, $M_t > 0$ for all t , and $p > 1$, it follows that $(M_t)_{t \geq 0}$ is uniformly integrable.

So, now we can let $t \rightarrow \infty$, and using the uniform integrability and Hölder's inequality for the exponents $\frac{1}{c}$ and $\frac{1}{1-c}$, by similar way as above, we have

$$1 = \mathbb{E} e^{cB_\tau - \frac{1}{2} c^2 \tau} \leq \left[\mathbb{E} e^{B_\tau - \frac{1}{2} \tau} \right]^c \left[\mathbb{E} e^{\frac{1}{2} \frac{c}{1-c} \tau} \right]^{1-c} \quad (3.7)$$

Taking $c \rightarrow 1$ in the above equation, equations 3.6 and 3.7 together provide us with our required result. $\square$

Remark 3.2.7. By the proof of the above theorem, we indeed have for all $c \in [0, 1]$, $\mathbb{E} e^{cB_\tau - \frac{1}{2} c^2 \tau} = 1$. Also, if we replace B by $-B$, we have this result for all $c \in [-1, 1]$. Note that, M_t is a martingale for all $c \in \mathbb{R}$, but the requirement of uniform integrability is necessary for the proof to go through, which is not available to us if we choose some $c \notin [-1, 1]$.

Chapter 4

Brownian motion as a Markov process

We have already seen the Markovian behaviour of a Brownian motion, that is, $(B_t)_{t \geq 0}$ is a Brownian motion, then, $B_t - B_s \perp \mathcal{F}_s$ for all $s \leq t$. Now, as in the renewal property of BM, we define $W_t := B_{t+s} - B_s$, then considering the canonical filtrations of B and W respectively, we have $\mathcal{F}_s^B \perp \mathcal{F}_\infty^W := \sigma(B_{t+s} - B_s : t \geq 0)$. The following theorem now quite intuitively follow from the definition of admissible filtrations.

Theorem 4.0.1. *Let $(B_t)_{t \geq 0}$ be a BM^d and $(\mathcal{F}_t)_{t \geq 0}$ be an admissible filtration. Then for every $s > 0$, the process W_t , $t \geq 0$ is also a BM^d and $(W_t)_{t \geq 0}$ is independent of $\mathcal{F}_s$, that is, $\mathcal{F}_\infty^W \perp \mathcal{F}_s$.*

The above theorem simply implies that we can always split up a Brownian motion into two independent pieces:

$$B_{t+s} = W_t + y|_{y=B_s}.$$

Clearly, $W_t + y|_{y=B_s}$ is again a Brownian motion starting at $y \in \mathbb{R}^d$. For the sake of simplicity, let us now introduce some notations.

$$\mathbb{P}^x(B_{t_1} \in A_1, \dots, B_{t_n} \in A_n) := \mathbb{P}(B_{t_1} + x \in A_1, \dots, B_{t_n} + x \in A_n), \quad (4.1)$$

where $0 \leq t_1 < \dots < t_n$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}^d)$. We can also analogously define the expectation $\mathbb{E}^x$ for the corresponding mathematical expectation. Here, $\mathbb{P}^x(B_s \in A)$ denotes the probability that the Brownian motion starts at the point x and traverses s units of time into the set A . By virtue of equation 4.1 and definition of $\mathbb{P}^x$, it is clear that $\mathbb{P}^x$ is a unique and well defined measure on $(\Omega, \mathcal{F}_\infty)$. Also, $x \mapsto \mathbb{E}^x u(B_t) = \mathbb{E}^0 u(B_t + x)$ is for all $u \in \mathcal{B}(\mathbb{R}^d)$ a measurable function.

Now, to proceed with our discussion in this chapter, we will need a very important result from conditional expectations.

Lemma 4.0.2. *If $\mathcal{G}$ and $\mathcal{H}$ are two independent sub- σ algebras of $\mathcal{A}$, and X and Y are $\mathcal{G}/\mathcal{B}(\mathbb{R}^d)$ and $\mathcal{H}/\mathcal{B}(\mathbb{R}^d)$ measurable random variables, then for all bounded Borel measurable functions $\Phi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, we have*

$$\mathbb{E} [\Phi(X, Y) | \mathcal{G}] = \mathbb{E} [\Phi(x, Y) |_{x=X}] = \mathbb{E} [\Phi(X, Y) | X]. \quad (4.2)$$

If $\psi : \mathbb{R}^d \times \Omega \rightarrow \mathbb{R}$ is bounded and $(\mathcal{B}(\mathbb{R}^d) \otimes \mathcal{H})/\mathcal{B}(\mathbb{R})$ measurable, then

$$\mathbb{E} [\Psi(X(\cdot), \cdot) | \mathcal{G}] = \mathbb{E} [\Psi(x, \cdot) |_{x=X}] = \mathbb{E} [\Psi(X(\cdot), \cdot) | X]. \quad (4.3)$$

This result seems intuitively clear from the properties of conditional expectations, but one does obviously need a formal proof, which we skip, but following are some nice applications of this lemma.

Theorem 4.0.3 (Markov property). *Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$. Then*

$$\mathbb{E} [u(B_{t+s}) | \mathcal{F}_s] = \mathbb{E} [u(B_t + x) |_{x=B_s}] = \mathbb{E}^{B_s} [u(B_t)] \quad (4.4)$$

holds for all bounded measurable functions $u : \mathbb{R}^d \rightarrow \mathbb{R}$, and

$$\mathbb{E} [\Psi(B_{\cdot+s}|\mathcal{F}_s)] = \mathbb{E} [\Psi(B_{\cdot} + x)]|_{x=B_s} = \mathbb{E}^{B_s} [\Psi(B_{\cdot})] \quad (4.5)$$

holds for all bounded $\mathcal{B}(\mathcal{C})/\mathcal{B}(\mathbb{R})$ measurable functions $\Psi : \mathcal{C}[0, \infty) \rightarrow \mathbb{R}$ which may depend upon the whole Brownian path.

Proof. Equation 4.4 follows from 4.2 by taking $X = B_s$, $Y = B_{t+s} - B_s$, $\mathcal{G} = \mathcal{F}_s$ and the function $\Phi(X, Y) = u(x + y)$.

Equation 4.5 follows from 4.5 if we take $\mathcal{H} = \mathcal{F}_\infty^W$, where $W_t = B_{t+s} - B_s$ and $\mathcal{G}, X$ are same as above. $\square$

Remark 4.0.4. 1. If observed closely, one might decipher that theorem 4.0.1 is not only restricted to Brownian motions, but it holds true for any stationary and independent stochastic process. Also, on the other hand, lemma 4.0.2 holds in even more generality than this, as any general d -dimensional, $\mathcal{F}_t$ adapted stochastic process $(X_t)_{t \geq 0}$ with right continuous paths is called a Markov process if it satisfies

$$\mathbb{E} [u(X_{t+s})|\mathcal{F}_s] = \mathbb{E} [u(X_{t+s})|X_s] \quad (4.6)$$

for all $s, t \geq 0$ and $u \in \mathcal{B}_b(\mathbb{R}^d)$.

2. We can always select a unique Borel measurable function $g_{u,s,t}(X_s)$ to express $\mathbb{E} [u(X_{t+s})|X_s]$ in terms of X_s . Now, if the function $g_{u,s,t+s}$ depends only on the difference $(t+s) - s = t$, then our Markov process in consideration is a *homogeneous Markov process*, for which we have

$$\mathbb{E} [u(X_{t+s})|\mathcal{F}_s] = \mathbb{E}^{X_s} [u(X_t)]$$

with $\mathbb{E}^x [u(X_t)] := g_{u,t}(x)$.

3. It is often useful to write equation 4.6 in its integral form, that is, for all ω ,

$$\begin{aligned} \mathbb{E} [u(B_{t+s}|\mathcal{F}_s)](\omega) &= \int_{\Omega} u(B_t(\omega') + B_s(\omega)) \mathbb{P}(d\omega') \\ &= \int_{\Omega} u(B_t(\omega')) \mathbb{P}^{B_s(\omega)}(d\omega') = \mathbb{E}^{B_s(\omega)} [u(B_t)]. \end{aligned}$$

4. In the case of discrete Markov processes, we know how Chapman-Kolmogorov equations help in determining transition probabilities between any two states. Here, with an analogous viewpoint, we can use equation 4.4 to obtain finite dimensional distributions of Brownian motion $(b_t)_{t \geq 0}$ from one dimensional distributions. Let us elaborate more on this.

Consider time points $0 \leq s < t < u$ and real valued bounded measurable functions ϕ, ψ, χ . Then,

$$\begin{aligned} \mathbb{E} [\phi(B_s) \psi(B_t) \chi(B_u)] &= \mathbb{E} [\phi(B_s) \psi(B_t) \mathbb{E} [\chi(B_u)|\mathcal{F}_t]] \\ &= \mathbb{E} [\phi(B_s) \psi(B_t) \mathbb{E}^{B_t} [\chi(B_{u-t})]] \\ &= \mathbb{E} [\phi(B_s) \mathbb{E}^{B_s} [\psi(B_{t-s})] \mathbb{E}^{B_t} [\chi(B_{u-t})]] \\ &= \mathbb{E} [\phi(B_s) \mathbb{E}^{B_s} ([\psi(B_{t-s})] \mathbb{E}^{B_{t-s}} [\chi(B_{u-t})])] \end{aligned}$$

where in the 1-st line we have used the fact that B_u is $\mathcal{F}_t$ measurable which is true since $u > t$, and in the 3rd and 4th lines we have used equation 4.4.

Now, we have

$$\mathbb{E}^{B_s}([\psi(B_{t-s})]\mathbb{E}^{B_{t-s}}[\chi(B_{u-t})]) = \int \psi(a) \int \xi(b) \mathbb{P}^a(\chi_{u-t} \in db) \mathbb{P}^{B_s}(B_{t-s} \in da),$$

and after rewriting all the expectations in the above expressions in their integral forms, we obtain

$$\begin{aligned} \int \int \int \phi(x) \psi(y) \chi(z) \mathbb{P}(B_s \in dx, B_t \in dy, B_u \in dz) = \\ \int \left[\phi(c) \int \psi(a) \int \xi(b) \mathbb{P}(\chi_{u-t} \in db) \mathbb{P}(B_{t-s} \in da) \right] \mathbb{P}(B_s \in dc). \end{aligned}$$

This shows that if we want to know the joint distribution (B_s, B_t, B_u) , then it is enough to know the one dimensional distributions B_t for each t .

If we put $s = 0$, and $\phi(0) = 1$ in the above equations, then the relations that we obtain are also called the *Chapman-Kolmogorov equations*. One can see that it's the same Chapman-Kolmogorov equations that we know if we just consider the probability distributions in place of expectations. The following theorem is what we mean by this (we have this result for any general Markov process by virtue of equation 4.2).

Theorem 4.0.5. *Let $(B_t)_{t \geq 0}$ be a BM^d or a general Markov process, $x_0 = 0$, $t_0 = 0 < t_1 < \dots < t_n$. Then*

$$\mathbb{P}^0(B_{t_1} \in dx_1, \dots, B_{t_n} \in dx_n) = \prod_{j=1}^n \mathbb{P}^{x_{j-1}}(B_{t_j - t_{j-1}} \in dx_j).$$

Till now, we have only seen Brownian motion stopping at any arbitrary time point s , and then continuing its journey from there which gives us a renewed Brownian motion $B_{t+s} - B_s$. But, the interesting part is that, if we replace any arbitrary point s with a stopping time, say σ , which is measurable with respect to any admissible filtration of the Brownian motion, then the renewed process $B_{t+\sigma} - B_\sigma$ is again a BM, which has a Markovian property stronger than what we have seen till now! We call this to be the strong Markov property of a BM, and our next section discusses this in detail.

4.1 The strong Markov property

If $(X_t)_{t \geq 0}$ is a stochastic process and σ is a stopping time, we set

$$X_\sigma(\omega) := \begin{cases} X_{\sigma(\omega)}(\omega) & \text{if } \sigma(\omega) < \infty, \\ \lim_{t \rightarrow \infty} X_t(\omega) & \text{if } \sigma(\omega) = \infty, \\ 0 & \text{in all other cases.} \end{cases}$$

This setting is required because σ can even take the value ∞ , so we have to ensure that B_σ is defined in this case.

The following theorem tells that equation 4.4 still remains valid even when we replace the point s with a stopping time $\sigma(\omega)$.

Theorem 4.1.1 (Strong Markov property). *Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$ and $\sigma < \infty$ be a.s. finite stopping time. Then $(W_t)_{t \geq 0}$ defined by $W_t := B_{t+\sigma} - B_\sigma$ is again a BM^d , which is independent of $\mathcal{F}_{\sigma+}$.*

By virtue of the above theorem and equations 4.2 and 4.3, the following theorem is immediate.

Theorem 4.1.2 (Markov property corresponding to stopping time). *Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$, and let σ be a stopping time. Then*

$$\mathbb{E} [u(B_{t+\sigma}) | \mathcal{F}_{\sigma+}] = \mathbb{E} [u(B_t + x)] |_{x=B_\sigma(\omega)} = \mathbb{E}^{B_\sigma(\omega)} [u(B_t)] \quad (4.7)$$

holds for all bounded measurable functions $u : \mathbb{R}^d \rightarrow \mathbb{R}$, and $\mathbb{P}$ almost all $\omega \in \{\sigma < \infty\}$. Also,

$$\mathbb{E} [\Psi(B_{\cdot+\sigma} | \mathcal{F}_s)] = \mathbb{E} [\Psi(B_{\cdot} + x)] |_{x=B_\sigma} = \mathbb{E}^{B_\sigma} [\Psi(B_{\cdot})] \quad (4.8)$$

holds for all bounded $\mathcal{B}(\mathcal{C})/\mathcal{B}(\mathbb{R})$ measurable functions $\Psi : \mathcal{C}[0, \infty) \rightarrow \mathbb{R}$ (which may depend upon the whole Brownian path) and $\mathbb{P}$ almost all $\omega \in \{\sigma < \infty\}$.

After this it is obvious to wonder what happens if we now replace both s and t with stopping times. Will the expression in 4.4 still remain valid? Well, for some particular stopping times, the answer is a big yes! Not only that, it in fact comes off as a corollary to the above theorem.

Let $\tau(\omega) = \tau(B_{\cdot}(\omega))$ be a stopping time which is expressed as a functional of a Brownian path, for example, a first hitting time. If σ is a further stopping time such that $\sigma \leq \tau$ a.s., then we can define a stopping time $\tau' = \tau(B_{\cdot+\sigma})$, which is the stopping time τ for the shifted process $B_{\cdot+\sigma}$, that is, it is the remaining time counting from σ until the event described by τ happens. Now, if we set $W_{\cdot} := B_{\cdot+\sigma} - B_\sigma$, the stopping time τ' can now be expressed as $\tau' = \tau(W_{\cdot} + B_\sigma)$ and note that for any $u \in \mathcal{B}_b(\mathbb{R}^d)$, the functional $u(B_\tau)$ and $u(W_{\tau'} + B_\sigma)$ have the same distribution, hence the below corollary follows from 4.8.

Corollary 4.1.3. *Let $(B_t)_{t \geq 0}$ be a BM^d , and $\tau = \tau(B_{\cdot})$ a first hitting time and σ a stopping time such that $\sigma \leq \tau$ a.s. Set $\tau' = \tau(B_{\cdot+\sigma})$ and $W_{\cdot} := B_{\cdot+\sigma} - B_\sigma$. Then*

$$\mathbb{E} [u(B_\tau) | \mathcal{F}_{\sigma+}] = \mathbb{E} [u(W_{\tau'} + x)] |_{x=B_\sigma(\omega)} = \mathbb{E}^{B_\sigma(\omega)} [u(W_{\tau'})]$$

holds for all $u \in \mathcal{B}_b(\mathbb{R}^d)$ and $\mathbb{P}$ almost all $\omega \in \{\tau < \infty\}$.

Remark 4.1.4 (Strong Markov process). Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process which is $\mathcal{F}_t$ adapted and has right continuous paths. If $(X_t)_{t \geq 0}$ satisfies for all $u \in \mathcal{B}_b(\mathbb{R}^d)$,

$$\mathbb{E} [u(X_{\sigma+t}) | \mathcal{F}_{\sigma+}] = \mathbb{E} [u(X_{\sigma+t}) | X_\sigma]$$

on $\{\sigma < \infty\}$, is called a *strong Markov process*.

The following section discusses about a principle which is one of the most famous applications of the strong Markov property of a Brownian motion.

4.2 Desiré André's reflection principle

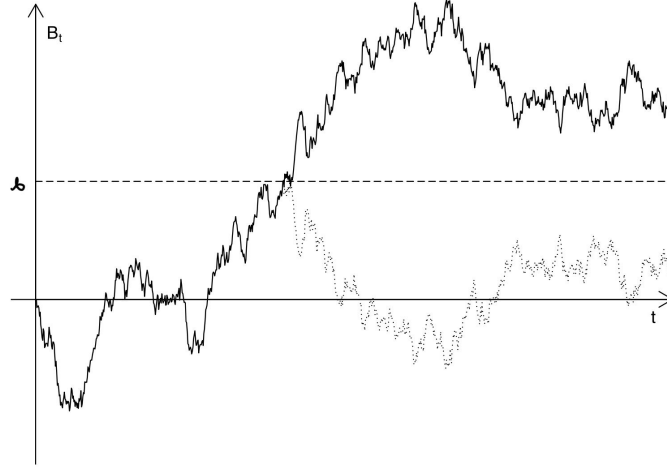
For BM^1 $(B_t)_{t \geq 0}$, let $\tau_b = \inf\{t \geq 0 : B_t = b\}$ be the first passage time for the level b . Now, since,

$$W_s + b = B_{\tau_b+s} - B_{\tau_b} + b = B_{\tau_b+s}$$

is again a BM, we have, $\mathbb{P}(B_t \geq b) = \mathbb{P}(B_t < b)$, that is, along the axis $B_{\tau_b} = b$, the Brownian motion gets reflected.

Now, we can write

$$\mathbb{P}(\tau_b \leq t, B_t < b) = \mathbb{P}(\{\tau_b \leq t\} \cap \{B_{\tau_b+(t-\tau_b)} - B_{\tau_b} < 0\}) = \frac{1}{2} \mathbb{P}(\tau_b \leq t),$$

Figure 4.1: Reflection along the axis $B_{\tau_b} = b$.

since, $\{\tau_b \leq t\} \in \mathcal{F}_{\tau_b}^B$ and $\{B_{\tau_b+(t-\tau_b)} - B_{\tau_b} < 0\} \in \mathcal{F}_{\infty}^W \amalg \mathcal{F}_{\tau_b}^B$. Hence,

$$\begin{aligned} \mathbb{P}(\tau_b \leq t) &= \mathbb{P}(\tau_b \leq t, B_t \geq b) + \mathbb{P}(\tau_b \leq t, B_t < b) \\ &= \mathbb{P}(\tau_b \leq t, B_t \geq b) + \frac{1}{2}\mathbb{P}(\tau_b \leq t) \\ &= \mathbb{P}(B_t \geq b) + \frac{1}{2}\mathbb{P}(\tau_b \leq t). \end{aligned}$$

This shows that

$$\mathbb{P}(\tau_b \leq t) = 2\mathbb{P}(B_t \geq b).$$

Henceforth, defining $M_t := \sup_{s \leq t} B_s$, and observing that $\{\tau_b \leq t\} = \{M_t \geq b\}$, we have,

$$\mathbb{P}(M_t \geq b) = \mathbb{P}(\tau_b \leq t) = 2\mathbb{P}(B_t \geq b) \text{ for all } b > 0.$$

4.2.1 Probability distributions of various functionals of a BM

The Desiré André's reflection principle allows us to calculate the probability distributions of the following important functionals of a BM.

Theorem 4.2.1. *Let $(B_t)_{t \geq 0}$ be a BM^1 , $b \in \mathbb{R}$, τ_b and M_t as defined before. Furthermore, if $m_t = \inf_{s \leq t} B_s$, then, for all $x \geq 0$,*

$$M_t \sim |B_t| \sim -m_t \sim M_t - B_t \sim B_t - m_t \sim \sqrt{\frac{2}{\pi t}} e^{x^2/(2t)} dx, \quad (4.9)$$

and for $t > 0$,

$$\tau_b \sim \frac{|b|}{\sqrt{2\pi t^3}} e^{-b^2/2t} dt. \quad (4.10)$$

Proof. From equation ??, it follows that $M_t \sim |B_t|$. Also, using the properties such as time reversal and reflection of a BM, the rest of the equivalences in 4.9 follows. Equation 4.9, for $b > 0$ can also be achieved by using the fact $\{M_t \geq b\} = \{\tau_b \leq t\}$ and differentiating their probability distribution. For $b < 0$, again the symmetric nature of the BM is used. $\square$

Remark 4.2.2. Note that equation 4.9 only tells that the one dimensional distributions of the random variables, example, M_t and B_t are same, and *not* the finite dimensional distributions! It is easy to see that it can never be the case that the laws of these two processes are same, because at one hand $t \rightarrow M_t$ is a monotonically increasing function, but on the other hand, $t \rightarrow |B_t|$, although positive, is never monotonic. But, the processes m_t and M_t are equivalent, that is, they do have the same law.

4.2.2 A more stronger version of the strong Markov property

Now, let us look at a more stronger version of our strong Markov property, which vaguely states that in corollary 4.3.3, we can choose σ to be any $\mathcal{F}_t^B$ stopping time and τ to be any $\mathcal{F}_{\sigma+}^B$ random time.

Theorem 4.2.3. *Let $(B_t)_{t \geq 0}$ be a BM^d , τ be an $\mathcal{F}_t^B$ measurable stopping time, and $\eta \geq \tau$ be a $\mathcal{F}_{\tau+}^B$ measurable random time. Then for all $u \in \mathcal{B}_b(\mathbb{R}^d)$, and $\omega \in \{\eta < \infty\}$,*

$$\begin{aligned} \mathbb{E} [u(B_\eta) | \mathcal{F}_{\tau+}^B](\omega) &= \mathbb{E}^{B_\tau(\omega)} [u(B_{\eta(\omega)-\tau(\omega)}(\cdot))] \\ &= \int u(B_{\eta(\omega)-\tau(\omega)}(\omega')) \mathbb{P}^{B_\tau(\omega)}(d\omega'). \end{aligned}$$

Remark 4.2.4. In the above theorem, the dependence of the random times on the fixed path ω should be noted. Also, some examples of the (τ, η) pairs are $(\tau, \tau + t)$, $(\tau, \tau \wedge t)$, $(\sigma \vee t, \sigma)$, where σ is a further stopping time.

Proof. If we replace $u(B_\eta)$ by $u(B_\eta)\mathbb{1}_{\{\eta < \infty\}}$, then, we can assume that $\eta < \infty$. Now, in the expression $\mathbb{E} u(B_\eta)$, we need to write η in a way so that we can use the strong Markov property of the BM. For that, let us consider

$$\sigma_j := \frac{\lfloor 2^j(\eta - \tau) \rfloor + 1}{2^j}.$$

Then, note that $\inf_{j \geq 1} \sigma_j = \eta - \tau$ and each σ_j is $\mathcal{F}_{\tau+}^B$ measurable, since η and τ are $\mathcal{F}_{\tau+}^B$ measurable. Since, the numerator of σ_j is always integer, and for all $F \in \mathcal{F}_{\tau+}^B$, we have $F \cap \{\sigma_j = k/2^j\} \in \mathcal{F}_{\tau+}^B$, we have

$$\begin{aligned} \int_F u(B_{\tau+\sigma_j}) d\mathbb{P} &= \sum_{k=1}^{\infty} \int_{F \cap \{\sigma_j = k/2^j\}} u(B_{\tau+k/2^j}) d\mathbb{P} \\ &= \sum_{k=1}^{\infty} \int_{F \cap \{\sigma_j = k/2^j\}} \mathbb{E} [u(B_{\tau+k/2^j}) | \mathcal{F}_{\tau+}^B] d\mathbb{P} \\ &= \sum_{k=1}^{\infty} \int_{F \cap \{\sigma_j = k/2^j\}} \mathbb{E}^{B_\tau(\omega)} [u(B_{k/2^j})] d\mathbb{P} \\ &= \int_F \mathbb{E}^{B_\tau(\omega)} [u(B_{k/2^j})] \mathbb{P}(d\omega'). \end{aligned}$$

Since, $t \rightarrow B_t$ is a continuous map, as $j \rightarrow \infty$, by DCT on both sides of the above integrals, we have our result for all $u \in \mathcal{C}_l(\mathbb{R}^d)$. Now, to generalize this to the entire class of bounded Borel measurable functions, we use the fact that for any compact set $K \subset \mathbb{R}^d$, the characteristic function $\mathbb{1}_K$ can be approximated from above by a sequence of continuous functions. Hence using MCT in this case, we have

$$\int_F u(B_\eta) d\mathbb{P} = \int_F \mathbb{E}^{B_\tau(\omega)} [u(B_{\eta(\omega)-\tau(\omega)})] \mathbb{P}(d\omega').$$

Now, as the compact sets generate the entire Borel- σ algebra, we can use the uniqueness theorem of measures to extend the above equality to all functions of the form $\mathbb{1}_K$, where $K \in \mathcal{B}(\mathbb{R}^d)$. Then, we have our result for all step functions, and finally, for all $u \in \mathcal{B}_b(\mathbb{R}^d)$. $\square$

4.2.3 General version of the reflection principle

The reflection principle just makes use of the strong Markov property of a BM, and the symmetry of its distribution. So, the this principle can be generalised as follows:

Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a Brownian motion and τ be an a.s. finite stopping time. Then the process

$$W(t, \omega) := \begin{cases} B(t, \omega) & \text{if } t \leq \tau(\omega) \leq \infty, \\ 2B(\tau(\omega), \omega) - B(t, \omega) & \text{if } \tau(\omega) \leq t < \infty. \end{cases} \quad (4.11)$$

which is the concatenation of the stopped process $(B(\tau \wedge t))_{t \geq 0}$ and $(B(\tau) - B(\tau + t))$ (which is independent of the former stopped process) is again a BM. [Note that the process W coincides with B in the interval $[0, \tau(\omega))$, and in the interval $[\tau(\omega), \infty)$, it is just the reflection of the process B about the axis $y = B(\tau(\omega), \omega)$.

Lemma 4.2.5. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM¹ and τ be an a.s. finite stopping time. Then $B(\tau \wedge t)$ is $\mathcal{F}_{\tau+}$ measurable for every $t \geq 0$.*

4.3 Transience and recurrence

To start with this section, we first need some general results. We have previously seen some examples of martingales, submartingales and supermartingales, all of which were of the form $f(B_t)$, for some function f . We will be able to generate many more martingales of this kind by using the following results.

Lemma 4.3.1. *Let $p(t, x) := (2\pi t)^{-d/2} e^{-|x|^2/2t}$ be the transition density of a d -dimensional Brownian motion. Then, for all $x \in \mathbb{R}^d$ and $t > 0$,*

$$\frac{\partial}{\partial t} p(t, x) = \frac{1}{2} \Delta_x p(t, x), \quad (4.12)$$

where Δ_x is the d -dimensional Laplace operator. Moreover,

$$\int p(t, x) \frac{1}{2} \Delta_x f(t, x) dx = \int f(t, x) \frac{1}{2} \Delta_x p(t, x) dx = \int f(t, x) \frac{\partial}{\partial t} p(t, x) dx, \quad (4.13)$$

for all $f \in \mathcal{C}^{1,2}((0, \infty) \times \mathbb{R}^d) \cap \mathcal{C}([0, \infty) \times \mathbb{R}^d)$ satisfying

$$|f(t, x)| + \left| \frac{\partial f(t, x)}{\partial t} \right| + \sum_{j=1}^d \left| \frac{\partial f(t, x)}{\partial x_j} \right| + \sum_{j,k=1}^d \left| \frac{\partial^2 f(t, x)}{\partial x_j \partial x_k} \right| \leq c(t) e^{C|x|}, \quad (4.14)$$

for all $t > 0$, some constant $C > 0$ and with a locally bounded function $c : (0, \infty) \rightarrow [0, \infty)$.

Proof. Equation 4.12 is just a routine check and equation 4.13 is obtained after applying by parts two times. $\square$

Note that the inequality 4.14 ensures that all the marginal terms in the previous inequalities vanish, that is, all the integrals exist.

Theorem 4.3.2. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^d and assume that $f \in \mathcal{C}^{1,2}((0, \infty) \times \mathbb{R}^d) \cap \mathcal{C}([0, \infty) \times \mathbb{R}^d)$ satisfies equation 4.14. Then, for $t \geq 0$,*

$$M_t^f := f(t, B_t) - f(0, B_0) - \int_0^t Lf(r, B_r) dr \quad (4.15)$$

is an $\mathcal{F}_t$ martingale, where, $Lf(t, x) = \frac{\partial}{\partial t} f(t, x) + \frac{1}{2} \Delta_x f(t, x)$.

Now, let us get back to our discussion of the transience and recurrence properties. Corollary 3.2.3 shows the probability of the first exit time from an open interval of the form $(-a, b)$, where $a < b$, $b \in \mathbb{R}$. This result can also be extended to the case of continuous martingales.

Corollary 4.3.3. *Let $(M_t^x, \mathcal{F}_t)_{t \geq 0}$ be a real-valued martingale with continuous paths and $M_0 = x \in \mathbb{R}$ a.s. If the first exit time $\tau = \tau_{(r,R)^c}$ from the interval (r, R) , $r < R$ is a.s. finite, then we have for all $x \in \mathbb{R}$,*

$$\mathbb{P}(M_\tau^x = r) = \frac{R - x}{R - r}, \quad \text{and} \quad \mathbb{P}(M_\tau^x = R) = \frac{x - r}{R - r}.$$

Using a suitable function f , combining theorem 4.3.2 and corollary 4.3.3, the exit probabilities of a BM^1 from an annulus $B(0, R) \setminus B(0, r)$ can be obtained. The following theorem improves upon this observation by giving a more interesting result for any d -dimensional BM.

Theorem 4.3.4. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a Brownian motion with right continuous filtration, $r < R$, and denote $\tau_{B(0,r)}$ and $\tau_{B^c(0,R)}$ be the first hitting times of the set $B(0, r)$ and $B^c(0, R)$. Then, for all $r < |x| < R$,*

$$\mathbb{P}^x(\tau_{B(0,r)} < \tau_{B^c(0,R)}) = \begin{cases} \frac{R-|x|}{R-r} & \text{if } d = 1, \\ \frac{\log R - \log |x|}{\log R - \log r} & \text{if } d = 2, \\ \frac{R^{2-d} - |x|^{2-d}}{R^{2-d} - r^{2-d}} & \text{if } d \geq 3. \end{cases} \quad (4.16)$$

The consequences of theorem 4.3.4 becomes especially interesting for a degenerate annuli, i.e. for $r \rightarrow 0$ and $R \rightarrow \infty$. Letting first $r \rightarrow 0$, and $R \rightarrow \infty$ yields

$$\mathbb{P}^x(\tau_{B(0,r)} < \tau_{B^c(0,R)}) \xrightarrow{r \rightarrow 0} \mathbb{P}^x(\tau_{\{0\}} \leq \tau_{B^c(0,R)}) \xrightarrow{R \rightarrow \infty} \mathbb{P}^x(\tau_{\{0\}} < \infty).$$

If we just restrict ourselves to $R \rightarrow \infty$, then, we get

$$\mathbb{P}^x(\tau_{B(0,r)} < \tau_{B^c(0,R)}) \xrightarrow{R \rightarrow \infty} \mathbb{P}^x(\tau_{B(0,r)} < \infty).$$

Corollary 4.3.5. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^d with a right-continuous filtration. Then, we have for all $|x| \geq r > 0$,*

$$\mathbb{P}^x(\tau_{\{0\}} < \infty) = \begin{cases} 1, & \text{if } d = 1, \\ 0, & \text{if } d \geq 2, \end{cases} \quad (4.17)$$

$$\mathbb{P}^x(\tau_{B(0,r)} < \infty) = \begin{cases} 1, & \text{if } d = 1, 2, \\ \left(\frac{|x|}{r}\right)^{2-d} & \text{if } d \geq 3. \end{cases} \quad (4.18)$$

By the above corollary, and the fact that a restarted Brownian motion is again a Brownian motion (by strong Markov property), we have the following important result:

Corollary 4.3.6. *Let $(B_t)_{t \geq 0}$ be a BM^d and $x \in \mathbb{R}^d$.*

1. *If $d = 1$, Brownian motion is point recurrent, that is,*

$$\mathbb{P}^x(B(t) = 0 \text{ infinitely often}) = 1.$$

2. *If $d = 2$, Brownian motion is neighbourhood recurrent, that is,*

$$\mathbb{P}^x(B(t) \in \mathbb{B}(0, r) \text{ infinitely often}) = 1 \text{ for every } r > 0.$$

3. *If $d \geq 3$, Brownian motion is transient, that is,*

$$\mathbb{P}^x(\lim_{t \rightarrow \infty} |B(t)| = \infty) = 1.$$

Chapter 5

The variation of Brownian paths

We are aware of the notion of functions of bounded variation. Now, these variations of functions has the following generalization, called *p-variation*: If $f : [0, \infty) \rightarrow \mathbb{R}^d$ is a non-random function and $\Pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$ a finite partition of the interval $[0, t]$, then, $|\Pi| = \max_{1 \leq j \leq n} (t_j - t_{j-1})$ is called the *fineness* or *mesh* of the partition, and for $p > 0$,

$$S_p^\Pi(f; t) := \sum_{t_{j-1}, t_j \in \Pi} |f(t_j) - f(t_{j-1})|^p = \sum_{j=1}^n |f(t_j) - f(t_{j-1})|^p$$

is called the *p-variation* sum for Π . Now, if we take the supremum over all the finite partitions, we obtain something called the *strong/total p-variation*:

$$\text{VAR}_p(f; t) = \sup\{S_p^\Pi(f; t) : \Pi \text{ is a finite partition of } [0, t]\}.$$

There is also a commonly used notion of *p-variation*: if $(\Pi_n)_{n \geq 1}$ is a sequence of finite partitions of $[0, t]$ such that $\lim_{n \rightarrow \infty} |\Pi_n| = 0$, then

$$\text{var}_p^\Pi(f; t) = \lim_{|\Pi_n| \rightarrow 0} S_p^{\Pi_n}(f; t).$$

Note that $\text{Var}_p^\Pi(f; t)$ is always defined, and if $\text{var}_p^\Pi(f; t)$ exists, then, $\text{Var}_p^\Pi(f; t) \geq \text{var}_p^\Pi(f; t)$.

Now, in the definition of the *p-variation sum*, the points t_j in the partition of $[0, t]$ are chosen to be real numbers between $[0, t]$. But, a nice fact is that it is enough to consider only rational numbers, or points from any dense subset of $[0, t]$ for the partition (which will simplify our calculations to a good extent), and still the definition of $\text{Var}_p^\Pi(f; t)$ will remain valid. Indeed, for every $\epsilon > 0$ and any such partition Π , we can always find some partition $\Pi' = \{0 = q_0 < q_1 < \dots < q_n = t\}$ of $[0, t]$ such that the points of the partition q_j belong to some dense subset of $[0, t]$, and

$$\sum_{j=1}^n |f(t_j) - f(t_{j-1})|^p < \epsilon, \tag{5.1}$$

and whence, it easily follows that $|S_p^\Pi(f; t) - S_p^{\Pi'}(f; t)| < c_{p,d}\epsilon$, where the constant $c_{p,d}$ depends only upon p and the dimension d . Finally, since

$$\text{VAR}_p^\Pi(f; t) = \sup_n \sup\{S_p^\Pi(f; t) : \Pi \text{ partition of } [0, t] \text{ with } \#\Pi = n\},$$

from 5.1, it follows that it is enough to consider rational points while calculating $\text{Var}_p^\Pi(f; t)$.

Now, in the next section, we will focus on the quadratic variation $\text{var}_2^\Pi(B(\cdot, \omega); t)$ of a Brownian motion.

5.1 Quadratic variation

In this section, it is enough to consider only 1-dimensional Brownian motions as the components of any d -dimensional Brownian motion are independent 1-dimensional Brownian motions, and it will not be difficult to generalize all these proofs to the d -dimensional case. The variation $\text{var}_p(f; t)$ might not exist for all p , and any random (or non-random) function, but the next theorem will show that $\text{var}_2(B; t)$ always exists, where $B(t)$ is a BM^1 , and usually it is called the quadratic variation of a Brownian motion.

5.1.1 L^2 -convergence

This subsection will focus on the L^2 -convergence of this quadratic variation.

Theorem 5.1.1. *Let $(B_t)_{t \geq 0}$ be a BM^1 and $(\Pi_n)_{n \geq 0}$ be a sequence of finite partitions of $[0, t]$ such that $\lim_{n \rightarrow \infty} |\Pi_n| = 0$. Then, the mean square limit exists:*

$$\text{var}_2(B; t) = L^2(\mathcal{P}) - \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t.$$

Proof. Let $\Pi = \{t_0 = 0 < t_1 < \dots < t_n\}$ be some partition of $[0, t]$. We have,

$$\mathbb{E} S_2^{\Pi}(B; t) = \sum_{j=1}^n \mathbb{E} [B(t_j) - B(t_{j-1})]^2 = t.$$

Now, we can use the properties like independent increments, stationary increments, scaling of Brownian motions and the fact that $|\Pi| \rightarrow 0$, to obtain the mean square limit:

$$\begin{aligned} \mathbb{E} [S_2^{\Pi}(B; t) - t]^2 &= \sum_{j=1}^n \mathbb{V}(B_{t_j} - B_{t_{j-1}})^2 \\ &= \sum_{j=1}^n \mathbb{E} [(B_{t_j} - B_{t_{j-1}})^2 - (t_j - t_{j-1})]^2 \\ &= \sum_{j=1}^n (t_j - t_{j-1})^2 \mathbb{E} (B(1)^2 - 1) \\ &\leq 2 \sum_{j=1}^n (t_j - t_{j-1})^2 \\ &= 2|\Pi|(t_j - t_{j-1}) = 2|\Pi|t \rightarrow 0. \end{aligned}$$

□

This theorem has the following important corollary which states that a.s. for every fixed ω , the sample paths of a Brownian motion are *not* a function of bounded/total variation.

Corollary 5.1.2. *Almost all Brownian paths are of infinite total variation. In fact, for all $p < 2$, $\text{Var}_p(B; t) = \infty$ a.s.*

Proof. For some $\delta > 0$, we consider $p = 2 - \delta$. Then, we can write

$$\sum_{j=1}^n (B(t_j) - B(t_{j-1}))^2 \leq \max_{t_j, t_{j-1} \in \Pi_n} |B(t_j) - B(t_{j-1})|^\delta \text{Var}_{2-\delta}(B; t).$$

By the L^2 -convergence obtained in the previous theorem, we have that there exists a subsequence for which the left hand side of the inequality converges a.s., and by the uniform continuity of the Brownian motion in the interval $[0, t]$, $\lim_{|\Pi_n| \rightarrow 0} \max_{t_j, t_{j-1} \in \Pi_n} |B(t_j) - B(t_{j-1})|^\delta = 0$, hence, $\text{Var}_{2-\delta}(B; t) = \infty$ a.s. □

Now, another analytic property of Brownian motion will follow going along the same lines of proof as above.

Definition 5.1.3 ((locally) Hölder continuous). A function $f : [0, \infty) \rightarrow \mathbb{R}$ is said to be (locally) Hölder continuous of order $\alpha > 0$ if for all compact intervals $[a, b] \subset [0, \infty)$, there exists a constant $c = c(f, \alpha, [a, b])$ such that

$$|f(t) - f(s)| \leq c|t - s|^\alpha \text{ for all } s, t \in [a, b].$$

Corollary 5.1.4. *Almost all Brownian paths are nowhere locally Hölder continuous for all $\alpha > \frac{1}{2}$.*

Theorem 5.1.1 tells about the L^2 -convergence of the quadratic variation of a Brownian path. The next section will show the a.s. convergence of a Brownian path, whose proofs are due to Lévy and Dudley.

5.1.2 Almost sure convergence

The first theorem is due to Lévy (1948) which shows that the almost sure convergence of the quadratic variation of Brownian paths is possible if the mesh size has $O(1/n)$. Later in 1973, Dudley produced another version to it saying that it is also enough to consider partitions of mesh size $o(1/\log n)$ for the respective a.s. convergence.

Theorem 5.1.5 (Lévy, 1948). *Let $(B_t)_{t \geq 0}$ be a BM^1 . For any sequence of finite partitions $(\Pi_n)_{n \geq 0}$ of $[0, t]$ which satisfies $\sum_{n=1}^{\infty} |\Pi_n| < \infty$, we have*

$$\text{var}_2(B; t) = \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t \text{ almost surely.}$$

Proof. In the proof of theorem 5.1.1, we have obtained

$$\mathbb{E} [(S_2^{\Pi_n} - t)^2] \leq 2|\Pi_n|t.$$

Now, using Markov-Chebyshev's inequality, we get

$$\sum_{j=1}^n \mathbb{P}(|S_2^{\Pi_n}(B; t) - t| > \epsilon) \leq \frac{2t}{\epsilon^2} \sum_{j=1}^n |\Pi_n| < \infty,$$

hence, the a.s. sure convergence is obtained using the 1st Borel-Cantelli lemma. $\square$

Note that $n|\Pi_n| \leq \sum_{j=1}^n |\Pi_j| \leq \sum_{j=1}^{\infty} |\Pi_j| < \infty$, hence, $|\Pi_n| = O(1/n)$.

Theorem 5.1.6 (Dudley, 1973). *Let $(B_t)_{t \geq 0}$ be a BM^1 and $|\Pi_n|$ be any sequence of partitions of $[0, t]$ which satisfies $|\Pi_n| = o(1/\log n)$. Then,*

$$\text{var}_2(B; t) = \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t \text{ almost surely.}$$

Proof. Since the proof of this theorem is quite big, we provide only the sketch of the proof.

Our given condition is that $\lim_{n \rightarrow \infty} \log n |\Pi_n| = 0$. In this proof, for simplification of notation, we will denote $S_2^{\Pi_n}(B; t)$ by $S_2^{\Pi_n}$. Now, we can write

$$S_2^{\Pi_n} = \sum_{j=1}^n s_j (W_j^2 - 1),$$

where $s_j = t_j - t_{j-1}$, and $W_j = \frac{B(t_j) - B(t_{j-1})}{\sqrt{t_j - t_{j-1}}}$. It is easy to see that for all $j = 1, \dots, n$, W_j are iid $N(0, 1)$ random variables. Now, for $\epsilon > 0$,

$$\mathbb{P}(|S_2^{\Pi_n} - t| > \epsilon) = \mathbb{P}(S_2^{\Pi_n} - t > \epsilon) + \mathbb{P}(t - S_2^{\Pi_n} > \epsilon).$$

Now, we claim that $\mathbb{P}(S_2^{\Pi_n} - t > \epsilon) \leq e^{\frac{-\epsilon\lambda_0}{2|\Pi_n|}}$. Suppose, the claim is true. Then, it follows that

$$\mathbb{P}(|S_2^{\Pi_n} - t| > \epsilon) \leq 2 \exp\left(\frac{-\epsilon\lambda_0}{2|\Pi_n|}\right).$$

Using the hypothesis, there exists a sequence ϵ_n of positive reals such that $|\Pi_n| = \frac{\epsilon_n}{\log n} \rightarrow 0$ as $n \rightarrow \infty$, and thus,

$$\mathbb{P}(|S_2^{\Pi_n} - t| > \epsilon) \leq 2n^{-\frac{\epsilon\lambda_0}{2\epsilon_n}} \leq \frac{2}{n^2},$$

whence, by 1st Borel-Cantelli lemma, the result follows. Now, to prove the claim, first using Markov inequality, we find

$$\mathbb{P}(|S_2^{\Pi_n} - t| > \epsilon) \leq e^{-\epsilon\theta} \mathbb{E}(e^{\theta(S_2^{\Pi_n} - t)}) = e^{-\epsilon\theta} \prod_{j=1}^n \mathbb{E}[e^{\theta s_j(W_j^2 - 1)}],$$

where the choice of the real no. θ is completely in our hands. This choice is made in such a way that we can ensure the right hand side of the inequality in our claim is a summable quantity, and hence the Borel-Cantelli lemma could be applied to obtain the required a.s. convergence. Now, using basic inequality $|e^{\lambda x} - \lambda x - 1| \leq \lambda^2 x^2 e^{|\lambda x|}$, $\lambda, x \in \mathbb{R}$ and using the fact that for each $j = 1, 2, \dots, n$, $\mathbb{E}[W_j^2 - 1] = 0$, it can be obtained

$$\mathbb{P}(|S_2^{\Pi_n} - t| > \epsilon) \leq e^{-\epsilon\theta} \prod_{j=1}^n \{1 + \mathbb{E}[\theta^2 s_j^2 (W_j^2 - 1)^2 e^{|\theta s_j| \cdot |W_j^2 - 1|}]\}.$$

Now, using the structure of a Gaussian random variable, in particular that the pdf contains the term $e^{-x^2/2}$, we can get $0 < \lambda_0 < \frac{1}{2}$ such that there exists a constant $C = C(\lambda_0)$ and

$$\sup_{\lambda \leq \lambda_0} \mathbb{E}[(W_j^2 - 1)^2 e^{|\lambda|(W_j^2 - 1)}] \leq C < \infty.$$

Plugging this into the previous expectation, the rest can be estimated by using the inequality $1 + y \leq e^y$ and choosing

$$\theta = \theta_0 = \min\left\{\frac{\epsilon}{2Cs^2}, \frac{\lambda_0}{|\Pi_n|}\right\},$$

where $s^2 = \sum_j s_j^2$. □

The next subsection shows yet another beautiful proof of a.s. convergence of the quadratic variation by Lévy using martingales.

5.1.3 Another proof of a.s. convergence using martingales

This subsection relies on the concept of backward martingales, the notion of closure of a martingale, and the backward convergence theorem for martingales. Let us first see what these are.

An adapted stochastic process $(X_t, \mathcal{F}_t)_{t \in I}$ is called a backward martingale if it is a martingale, but the index set is decreasing; that is, the first two properties of a martingale remains valid, only the third property replaced as follows:

$$\mathbb{E}[X_s | \mathcal{F}_t] = X_t, \text{ for all } s \leq t.$$

Theorem 5.1.7 (Closure of a martingale). *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a martingale in $\mathbb{R}^d$ with continuous paths. Then TFAE:*

1. $(X_t)_{t \geq 0}$ is uniformly integrable.
2. $\lim_{t \rightarrow \infty} X_t$ exists a.s. and in L^1 .

Theorem 5.1.8 (Backward convergence theorem). *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a martingale and let $t_1 \geq t_2 \geq \dots \geq t_n \geq \dots$ be a decreasing sequence of indices. Then, $(X_{t_j}, \mathcal{F}_{t_j})$ is a backwards martingale which is uniformly integrable. In particular, $\lim_{t \rightarrow \infty} X_t$ exists a.s. and in L^1 .*

From the above two theorems, we can conclude that backwards martingales are always uniformly integrable, and their limit always exists a.s. and in L^1 . Now, to prove the a.s. convergence by martingales, let us begin with a auxiliary lemma.

Lemma 5.1.9. *Let $(B_t)_{t \geq 0}$ be a BM¹ and $\mathcal{G}_s := \sigma(B_r : r \leq s, (B_u - B_v)^2 : u, v \geq s)$. Then,*

$$\mathbb{E} [B_t - B_s | \mathcal{G}_s] = 0, \text{ for all } 0 \leq s < t.$$

Proof. By definition of conditional expectation, it is enough to prove

$$\int_{\Gamma} B_t - B_s d\mathbb{P} = 0,$$

for all sets in the $\cap$ -stable generators, say Γ , of the sigma algebra $\mathcal{G}_s$. Such a typical set Γ will be of the form

$$\Gamma = \bigcap_{j=1}^M \{B_{r_j} \in C_{r_j}\} \cap \bigcap_{k=1}^N \{(B_{v_k} - B_{u_k})^2 \in D_k\}.$$

Now, using the facts $\mathcal{F}_s^B \amalg \mathcal{F}_{[s, \infty]}^B$ and that $\tilde{B} := -B$ is also a Brownian motion (reflection principle), it can easily be shown that

$$\int_{\Gamma} B_t - B_s d\mathbb{P} = - \int_{\Gamma} B_t - B_s d\mathbb{P},$$

and thus, the result follows. $\square$

Now, let us come to the actual theorem.

Theorem 5.1.10 (Lévy, 1940). *Let $(B_t)_{t \geq 0}$ be a BM¹ and $|\Pi_n|$ be a nested sequence of partition of $[0, t]$, that is, $\Pi_n \subset \Pi_{n+1} \subset \dots$, such that $\lim_{n \rightarrow \infty} |\Pi_n| = 0$. Then,*

$$\text{var}_2(B; t) = \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t \text{ almost surely.}$$

Proof. We can without loss of generality assume that $\#(\Pi_n \setminus \Pi_{n-1}) = 1$, as otherwise the missing partitions can be added. Also, we define

$$S_{-n} := S_2^{\Pi_n}(B; t), \text{ and } \mathcal{H}_{-n} := \sigma(S_{-n}, S_{-n-1}, \dots).$$

We show that S_{-n} is a backwards martingale with respect to the filtration $\mathcal{H}_{-n}$, because then by the backward convergence theorem, the a.s. convergence will follow. The first two properties of martingales are clear, hence it suffices to show

$$\mathbb{E} (S_{-n} - S_{-n-1} | \mathcal{H}_{-n-1}) = 0.$$

Let us suppose $\Pi_{n+1} = \Pi_n \cup \{s\}$. Then there exists points t_j and t_{j-1} in Π_n such that $t_{j-1} < s < t_j$. Then,

$$S_{-n} - S_{-n-1} = (B_{t_j} - B_s)^2 + (B_s - B_{t_{j-1}})^2 + (B_{t_j} - B_{t_{j-1}})^2 = 2(B_{t_j} - B_s)(B_s - B_{t_{j-1}})$$

and, since $B_s - B_{t_{j-1}}$ is $\mathcal{G}_s$ measurable,

$$\mathbb{E} [(B_{t_j} - B_s)(B_s - B_{t_{j-1}}) | \mathcal{G}_s] = (B_s - B_{t_{j-1}}) \mathbb{E} [B_{t_j} - B_s | \mathcal{G}_s],$$

whence using the previous lemma, the above quantity becomes 0. Now, as $\mathcal{H}_{n-1} \subset \mathcal{G}_s$, finally by tower property, we have

$$\mathbb{E} [S_n - S_{n-1} | \mathcal{H}_{n-1}] = \mathbb{E} [\mathbb{E} (S_n - S_{n-1} | \mathcal{G}_s) | \mathcal{H}_{n-1}] = 0.$$

By theorem 5.1.5, we know that the L^2 -limit is t , that is, there exists a subsequence of $S_2^{\Pi_n}$ that converges a.s. to t , but since we have proved the a.s. convergence of this entire sequence, we conclude that the a.s. limit of $S_2^{\Pi_n}$ is indeed t . $\square$

5.1.4 Almost sure divergence of the variance sum

So far we have only dealt with the deterministic model of the Brownian motion, and showed the quadratic variation converges a.s. and in L^2 . But, now we will see the behaviour of the variation paths when the partition Π depends upon ω . In this case, it was proven by Lévy that the quadratic variation diverges, and henceforth as a consequence, the strong quadratic variation $\text{VAR}_2(B(\cdot, \omega))$ also diverges. The dependence on ω indeed gives us a better picture of the variation in Brownian paths, and the reason is quite obvious: the quadratic variation that we saw in the deterministic approach cannot capture the actual variation of a particular Brownian path as succinctly as the dependence on ω would do.

Theorem 5.1.11 (Lévy, 1948). *Let $(B_t)_{t \geq 0}$ be a BM¹. For almost all ω , there exists a sequence of nested partitions $(\Pi_n(\omega))_{n \geq 1}$ such that $|\Pi_n(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, and*

$$\text{var}_2(B(\cdot, \omega); 1) = \lim_{n \rightarrow \infty} S_2^{\Pi_n(\omega)}(B(\cdot, \omega); 1) = \infty \quad \text{almost surely.}$$

Corollary 5.1.12. *Let $(B_t)_{t \geq 0}$ be a BM¹. Then, $\text{VAR}_2(B(\cdot, \omega); 1) = \infty$ for almost all $\omega \in \Omega$.*

5.2 Lévy's martingale characterization of Brownian motion

Theorem 5.1.1 is not only true for Brownian paths, but it also remains valid for continuous martingales $(X_t, \mathcal{F}_t)_{t \geq 0}$ with the property that $(X_t^2 - t, \mathcal{F}_t)_{t \geq 0}$ is also a martingale. It follows from the following observation, and the lemma after it.

Note that if $(X_t, \mathcal{F}_t)_{t \geq 0}$ is such a martingale as described above, then

$$\mathbb{E}[X_t - X_s | \mathcal{F}_s] = 0,$$

and,

$$\mathbb{E}[(X_t - X_s)^2 | \mathcal{F}_s] = t - s.$$

For the above equation, note that $\mathbb{E}[X_t X_s | \mathcal{F}_s] = X_s^2$, and thus, $\mathbb{E}[(X_t - X_s)^2 | \mathcal{F}_s] = \mathbb{E}[X_t^2 - X_s^2 | \mathcal{F}_s] = \mathbb{E}[(X_t^2 - t) - (X_s^2 - t) | \mathcal{F}_s] = (X_s^2 - s) - (X_s^2 - t) = t - s$, for all $s \leq t$.

Lemma 5.2.1. *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a continuous real valued martingale such that $(X_t^2 - t, \mathcal{F}_t)_{t \geq 0}$ is also a martingale. Then, $\mathbb{E}[X_t^4] < \infty$, and*

$$\mathbb{E}[(X_t - X_s)^4 | \mathcal{F}_s] \leq 4(t - s)^2.$$

Corollary 5.2.2. *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a martingale as described above. Then for all partitions $(\Pi_n)_{n \geq 1}$ of $[0, t]$ satisfying $\lim_{n \rightarrow \infty} |\Pi_n| = 0$,*

$$\text{var}_2(X; t) = L^2(\mathbb{P}) - \lim_{n \rightarrow \infty} S_2^{\Pi_n}(X; t) = t.$$

Till now, it might seem to anyone that this particular type of martingale satisfying the L^2 -convergence is just a matter of coincidence. But, in 1948 Lévy proved yet another result which stated that any martingale of such kind is actually a Brownian motion! This result is well known by the name of martingale characterization of a Brownian motion.

Theorem 5.2.3. *Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a real valued martingale with continuous sample paths such that $(X_t^2 - t, \mathcal{F}_t)_{t \geq 0}$ is also a martingale. Then, $(X_t)_{t \geq 0}$ is a Brownian motion.*

The following proof is due to Gikhman, Skorohod, 1972.

Proof. To proof this result, we use the characterization of Brownian motion via martingales that we have learnt earlier. We just need to show that for all $s \leq t$,

$$\mathbb{E} [e^{i\xi(X_t - X_s)}] = e^{-\frac{1}{2}(t-s)\xi^2}.$$

We will use Taylor's theorem (up to order 3) to prove this. To avoid cumbersome notations, we denote $\Delta X := X_t - X_s$ and $h = t - s$. Writing out the expressions, we obtain

$$e^{i\xi(X_t - X_s)} - e^{-i\xi^2(t-s)} = i\xi(\Delta X) + \frac{\xi^2}{2}h - \frac{\xi^2}{2}(\Delta X)^2 - \frac{\xi^4}{8}h^2 - \frac{i\xi^3}{6}(\eta\Delta X)^3 + \frac{\xi^6}{48}(\theta h)^3, \quad (5.2)$$

where $\eta = \eta(\omega) \in (0, 1)$, and $\theta \in (0, 1)$. Now, as discussed before, we have

$$\mathbb{E} [\Delta X | \mathcal{F}_s] = 0, \quad \text{and} \quad \mathbb{E} [(\Delta X)^2 | \mathcal{F}_s] = h.$$

Using the above and taking expectations both side of equation 5.2, we obtain

$$\mathbb{E} [e^{i\xi(X_t - X_s) - e^{-\frac{1}{2}\xi^2(t-s)}} | \mathcal{F}_s] = -\frac{\xi^4}{8}h^2 - \frac{i\xi^3}{6}\mathbb{E} [(\eta\Delta X)^3 | \mathcal{F}_s] + \frac{\xi^6}{48}(\theta h)^3.$$

Now, by the conditional version of the Hölder's inequality, where $p = 4/3$ and $q = 4$ are the conjugate exponents, we have

$$|\mathbb{E} [(\eta\Delta X)^3 | \mathcal{F}_s]| \leq \mathbb{E} [(\Delta X)^3 | \mathcal{F}_s] \leq \mathbb{E} [(\Delta X)^4 | \mathcal{F}_s]^{3/4} \leq 4^{3/4}h^{3/2},$$

where the last inequality follows using lemma 5.2.1. Hence, for $h < 1$ and some constant c_ξ ,

$$|\mathbb{E} [e^{i\xi(X_t - X_s) - e^{-\frac{1}{2}\xi^2(t-s)}} | \mathcal{F}_s]| \leq c_\xi(t-s)^{3/2}.$$

Now, for $s < t$, if we set $t_j := s + (t-s)\frac{j}{n}$, $j = 0, 1, \dots, n$, $n \geq 1$, and use the estimate

$$|\Pi_{j=1}^n a_j - \Pi_{j=1}^n b_j| \leq \sum_{j=1}^n |a_j - b_j|, \quad \text{for } a_j, b_j \in \mathbb{C}, |a_j|, |b_j| \leq 1,$$

we obtain

$$|\mathbb{E} [e^{i\xi(X_t - X_s) - e^{-\frac{1}{2}\xi^2(t-s)}} | \mathcal{F}_s]| \leq \sum_{j=1}^n \mathbb{E} [|e^{i\xi(X_{t_j} - X_{t_{j-1}}) - e^{-\frac{1}{2}\xi^2(t_j - t_{j-1})}} | \mathcal{F}_s].$$

Now, if $n \geq 1$ is very large, then, $h = t_j - t_{j-1}$ can be made < 1 for all $j = 0, 1, \dots, n$, and thus,

$$|\mathbb{E} [e^{i\xi(X_t - X_s) - e^{-\frac{1}{2}\xi^2(t-s)}} | \mathcal{F}_s]| \leq c_\xi \sum_{j=1}^n (t_j - t_{j-1})^{3/2} \leq \frac{c_\xi}{\sqrt{n}} \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

□

The above theorem gives way to the following corollary:

Corollary 5.2.4. *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^1 , where $(\mathcal{F}_t)_{t \geq 0}$ is an admissible filtration. Then, the stochastic process $(B_t^2 - t)_{t \geq 0}$ is a martingale with respect to $(\mathcal{F}_t)_{t \geq 0}$. Conversely, if $(X_t, \mathcal{F}_t)_{t \geq 0}$ is a real valued martingale with continuous paths such that $(X_t^2 - t)_{t \geq 0}$ is also a martingale with respect to $(\mathcal{F}_t)_{t \geq 0}$, then, $(X_t)_{t \geq 0}$ is a BM^1 .*

Chapter 6

Regularity of Brownian paths

Previously, we have defined the notion of Hölder continuity of a function and said that Brownian paths are nowhere Hölder continuous of any order $\alpha > 1/2$. In this chapter, the smoothness properties of Brownian paths will be discussed in detail, and we will show that the paths are in fact a.s. nowhere differentiable in t .

6.1 Hölder continuity

In this section, we will see the range of values for which a Brownian path (more precisely, one of its modification) is Hölder continuous for orders in that range.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be our probability space.

Theorem 6.1.1 (Kolmogorov 1934; Slutsky 1937; Chentsov 1956). *Let $(\xi(x))_{x \in \mathbb{R}^n}$ be a stochastic process on $(\Omega, \mathcal{A}, \mathbb{P})$ with values in $\mathbb{R}^d$ and index set $\mathbb{R}^n$. If*

$$\mathbb{E}[|\xi(x) - \xi(y)|^\alpha] \leq c|x - y|^{n+\beta}, \quad \text{for all } x, y \in \mathbb{R}^n, \quad (6.1)$$

holds for some constant $c > 0$, and $\alpha, \beta > 0$, then there exists a modification $(\xi'(x))_{x \in \mathbb{R}^n}$ with exclusively continuous sample paths. Moreover,

$$\mathbb{E} \left[\left(\sup_{\substack{0 < |x-y| < 1 \\ x, y \in [-T, T]^n}} \frac{|\xi(x) - \xi(y)|}{|x - y|^\gamma} \right)^\alpha \right] < \infty, \quad (6.2)$$

for all $T > 0$, and $0 \leq \gamma < \beta/\alpha$. In particular, $x \mapsto \xi'(x)$ is a.s. Hölder continuous for order γ .

Proof. Let $\mathbb{N}_0$ be the set of all natural numbers. Define $D_m := 2^{-m}\mathbb{N}_0 \cap [0, 1]^n$, and $D = \cup_{m \geq 0} D_m$, which thus contains all the dyadic numbers in $[0, 1]^n$. Next, we set

$$\Delta_m := \{(x, y) \in D_m \times D_m : |x - y|_\infty \leq 2^{-m}\},$$

where $|x - y|_\infty = \max_{1 \leq k \leq n} |x^k - y^k|$. Now, since we are in n -dimension, every point $x \in D_m$ has at most $3^n - 1$ nearest elements in D_m . Hence, the set Δ_m contains at most $3^n \cdot 2^{mn}$ elements. Now, we fix $\gamma \in (0, \beta/\alpha)$, and define $\sigma_j := \sup_{(x, y) \in \Delta_j} |\xi(x) - \xi(y)|$. Now, equation 6.1 shows that

$$\mathbb{E} [|\sigma_j|^\alpha] \leq \sum_{(x, y) \in \Delta_j} \mathbb{E} [|\xi(x) - \xi(y)|^\alpha] \leq c 3^n \cdot 2^{-j\beta}. \quad (6.3)$$

Now, consider any $x, y \in D$ such that $|x - y| < 2^{-m}$. To obtain the required result, we will be doing some sort of approximation of x, y using the dyadic numbers in $[0, 1]^n$. As per construction, for each $j \geq 0$, there exists unique elements $x_j, y_j \in D_j$ such that x and y are in the 2^{-j} cells $x_j + [0, 2^{-j})^n$ and $y_j + [0, 2^{-j})^n$ respectively. Now, clearly $\lim_{j \rightarrow \infty} x_j = x$ and $\lim_{j \rightarrow \infty} y_j = y$, and for $j = m$, the closed cells $x_m + [0, 2^{-m}]^n$ and $y_m + [0, 2^{-m}]^n$ has at least one element in common, let say, d_m . Thus, $(x_m, d_m), (d_m, y_m) \in \Delta_m$ and by triangle inequality,

$$|\xi(x_m) - \xi(y_m)| \leq 2\sigma_m.$$

Moreover, our construction shows that $(x_j, x_{j+1}), (y_j, y_{j+1}) \in \Delta_{j+1}$, for all $j \geq m$. Therefore,

$$\xi(x) - \xi(y) = \sum_{j \geq m} [\xi(x_{j+1}) - \xi(x_j)] + \xi(x_m) - \xi(y_m) - \sum_{j \geq m} [\xi(y_{j+1}) - \xi(y_j)],$$

hence, we get

$$|\xi(x) - \xi(y)| \leq 2\sigma_m + 2 \sum_{j \geq m+1} \sigma_j = 2 \sum_{j \geq m} \sigma_j.$$

So, using some careful analysis, it follows that

$$\sup_{\substack{x, y \in D \\ x \neq y}} \frac{|\xi(x) - \xi(y)|}{|x - y|^\gamma} = \sup_{m \geq 0} \sup_{\substack{x, y \in D \\ 2^{-(m+1)} < |x - y| < 2^{-m}}} \frac{|\xi(x) - \xi(y)|}{2^{-(m+1)\gamma}} \leq 2^{1+\gamma} \sum_{j=0}^{\infty} 2^{j\gamma} \sigma_j.$$

Now, for $\alpha \geq 1$, and $\gamma < \beta/\alpha$, we get

$$\begin{aligned} \left\| \sup_{\substack{x, y \in D \\ x \neq y}} \frac{|\xi(x) - \xi(y)|}{|x - y|^\gamma} \right\|_{l^\alpha(P)} &\leq 2^{1+\gamma} \sum_{j=0}^{\infty} 2^{j\gamma} \|\sigma_j\|_{L^\alpha(P)} \\ &\leq c^{1/\alpha} 2^{1+\gamma} 3^{n/\alpha} \sum_{j=0}^{\infty} 2^{j(\gamma - \beta/\alpha)} < \infty. \end{aligned}$$

For $\alpha < 1$, similar arguments work using $\mathbb{E} [|Z|^\alpha]$ instead of $\|Z\|_{L^\alpha(P)}$. Since, the expression inside the l^α -norm is a.s. finite, there exists a set $\Omega_1 \subset \Omega$ with $\mathbb{P}(\Omega_1) = 1$ such that for all $\omega \in \Omega_1$, we have

$$|\xi(x, \omega) - \xi(y, \omega)| \leq c(\omega) |x - y|^\gamma \text{ for all } x, y \in D,$$

that is, the paths $x \mapsto \xi(x, \omega)$ are uniformly continuous.

Now, this same construction can be easily generalised to intervals of the form $[-T, T]^n$, for all $T > 0$. We define $D^T = 2^{-m} \mathbb{N}_0 \cap [-T, T]^n$, and $D = \cup_{T \geq 0} D^T$. Also, as we obtained the probability space Ω_1 for the interval $[0, 1]^n$, we consider the probability spaces Ω_T for the intervals $[-T, T]^n$ for all $T > 0$. Hence, for all $T > 0$, $\mathbb{P}(\Omega_T) = 1$ and defining $\Omega_0 = \cap_{T \geq 0} \Omega_T$, we have $\mathbb{P}(\Omega_0) = 1$. So, for all $\omega \in \Omega_0$, the paths $x \mapsto \xi(x, \omega)$ are continuous for all $x \in D$. Now, finally we can do our required approximation for which we did all the former tedious constructions. Since, D is a dense set in $\mathbb{R}^n$, we set for all $x \in \mathbb{R}^n$,

$$\xi'(x, \omega) = \begin{cases} \lim_{D \ni y \rightarrow x} \xi(y, \omega) & \text{if } \omega \in \Omega_0 \\ 0 & \text{if } \omega \notin \Omega_0. \end{cases}$$

Let x_j be a sequence in D such that $x_j \rightarrow x$. By Fatou's lemma, and the hypothesis 6.1,

$$\begin{aligned} \mathbb{E} [|\xi(x) - \xi'(x)|^\alpha] &= \mathbb{E} [\liminf_{j \rightarrow \infty} |\xi(x) - \xi(x_j)|^\alpha] \\ &\leq \liminf_{j \rightarrow \infty} \mathbb{E} [|\xi(x) - \xi(x_j)|^\alpha] \\ &\leq c \liminf_{j \rightarrow \infty} |x - x_j|^{n+\beta} = 0. \end{aligned}$$

Therefore, $\mathbb{P}(\xi(x) = \xi'(x)) = 1$, for all $x, y \in \mathbb{R}^n$. Finally,

$$\begin{aligned} \sup_{\substack{0 < |x-y| < 1 \\ x, y \in [-T, T]^n}} \frac{|\xi'(x) - \xi'(y)|}{|x - y|^\gamma} &= \lim_{\substack{0 < |x-y| < 1 \\ x, y \in [-T, T]^n}} \lim_{\substack{D^T \ni u \rightarrow x \\ y \ni v \rightarrow y}} \frac{|\xi(u) - \xi(v)|}{|u - v|^\gamma} \\ &\leq \sup_{\substack{0 < |u-v| < 1 \\ x, y \in D}} \frac{|\xi(u) - \xi(v)|}{|u - v|^\gamma} \leq c_T(\omega), \end{aligned}$$

and this proves 6.2. Hence, it also follows that for all $x, y \in [-T, T]^n$ such that $|x - y| < 1$,

$$|\xi'(x) - \xi'(y)| \leq c(\omega)|x - y|^\gamma,$$

that is, the paths $x \rightarrow \xi'(x)$ are a.s. locally Hölder continuous of order γ . $\square$

Corollary 6.1.2. *A d -dimensional Brownian motion $(B_t)_{t \geq 0}$ is Hölder continuous up to order $\gamma < 1/2$.*

Proof. Since, any $\mathbb{R}^d$ valued Brownian motion is Hölder continuous iff its components are Hölder continuous, it is enough to consider $d = 1$. So, we have

$$\mathbb{E}[|B(t) - B(s)|^{2k}] = c_n |t - s|^k.$$

Now, we apply the previous theorem to obtain the result. (Note that since Brownian paths are already continuous, we need not consider any such modification of B_t). So, according to the previous theorem, we have $\alpha = 2k$, and $\beta = k - 1$. Then

$$\frac{\beta}{\alpha} = \frac{k - 1}{2k} \rightarrow \frac{1}{2} \text{ as } k \rightarrow \infty.$$

Now, following the proof of the above theorem, for each $\gamma < \frac{k-1}{2k}$, we have an exceptional set $N_k = \Omega \setminus \Omega_k$, where $\mathbb{P}(N_k) = 0$. Since, being Hölder continuous of order γ implies being Hölder continuous of order γ' , for all $\gamma' < \gamma$, the assertion for all $\gamma < \beta/\alpha$ follows by considering the null set $N = \cup_{k \geq 1} N_k$. $\square$

6.2 a.s. non-differentiability of Brownian paths

Now, we are finally at a stage where we can prove that for almost all $\omega \in \Omega$, the Brownian paths $t \mapsto B_t(\omega)$ is not differentiable. Due to the same reasons as described before, we can work with 1-dimensional BM. The proof of this result was first given by Payley, Weiner and Zygmund in 1931, but the proof that we give follows the classic exposition by Dvoretzky, Erdős, and Kakutani.

Theorem 6.2.1. *Let $(B_t)_{t \geq 0}$ be a BM¹. Then, the path $t \mapsto B_t(\omega)$ is for almost all $\omega \in \Omega$ nowhere differentiable.*

Proof. We consider the set

$$A_n := \{\omega \in \Omega : B(\cdot; \omega) \text{ is nowhere differentiable in } [0, n]\}.$$

Since, the set A_n may not be measurable, we cannot directly go on to proving $\mathbb{P}(A_n) = 1$, because this may not make sense. Instead, we prove that A_n^c is a null set, that is, there exists a measurable null set N_n such that $A_n^c \subset N_n$.

Now, for the interval $[0, n]$, for some $n \in \mathbb{N}$, if a Brownian path, say B_t is differentiable at a point $t_0 \in [0, n]$, then, there exists a $\delta > 0$ and $L > 0$ such that for all $t \in \mathbb{B}(t_0, \delta)$,

$$|B(t) - B(t_0)| \leq L|t - t_0|.$$

Now, for large enough

$$k \geq 1$$

, we can consider the grid $\{\frac{j}{k} : j = 1, 2, \dots, kn\}$, with respect to which there exists the smallest index $j = j(k)$ such that for $t_0 \leq \frac{j}{k}$, and $(j+1)/k, (j+2)/k, (j+3)/k \in \mathbb{B}(t_0, \delta)$. Now, by triangle inequality, and using the above definition of differentiability, it is easy to see that for $i = j+1, j+2, j+3$,

$$\left| B\left(\frac{i}{k}\right) - B\left(\frac{i-1}{k}\right) \right| \leq \frac{7L}{k}.$$

Henceforth, for sets of the form

$$C_m^L := \bigcap_{k=m}^{\infty} \bigcup_{j=1}^{kn} \bigcap_{i=j+1}^{j+3} \left\{ \left| B\left(\frac{i}{k}\right) - B\left(\frac{i-1}{k}\right) \right| \leq \frac{7L}{k} \right\},$$

we have

$$A_n^c \subset \bigcup_{L=1}^{\infty} \bigcup_{m=1}^{\infty} C_m^L.$$

Now, assertion will be proved if we can show that for all $L \geq 1$ and $m \geq 1$, $\mathbb{P}(C_m^L) = 1$. For $k \geq m$, using the properties of independent and stationary increments, we have

$$\begin{aligned} \mathbb{P}(C_m^L) &\leq \mathbb{P}\left(\bigcup_{j=1}^{kn} \bigcap_{i=j+1}^{j+3} \left\{ \left| B\left(\frac{i}{k}\right) - B\left(\frac{i-1}{k}\right) \right| \leq \frac{7L}{k} \right\}\right) \\ &\leq \sum_{j=1}^{kn} \mathbb{P}\left(\bigcap_{i=j+1}^{j+3} \left\{ \left| B\left(\frac{i}{k}\right) - B\left(\frac{i-1}{k}\right) \right| \leq \frac{7L}{k} \right\}\right) \\ &= \mathbb{P}\left(\left\{ \left| B\left(\frac{i}{k}\right) - B\left(\frac{i-1}{k}\right) \right| \leq \frac{7L}{k} \right\}\right)^3 \\ &= \sum_{j=1}^{kn} \mathbb{P}\left(\left| B\left(\frac{1}{k}\right) \right| \leq \frac{7L}{k}\right)^3 \\ &\leq kn \left(\frac{c}{\sqrt{k}}\right)^3 \rightarrow 0 \text{ as } k \rightarrow \infty, \end{aligned}$$

for some constant c . The last inequality follows from using the fact that $B\left(\frac{1}{k}\right) \sim \frac{1}{\sqrt{k}}B(1)$, and directly calculating the estimate. $\square$

We have finally proved the non-differentiability of Brownian paths! Hence, other than Weierstrass function we have obtained yet another function, our Brownian paths, which, for almost all fixed $\omega \in \Omega$, are continuous but nowhere differentiable.

We finish this chapter by completely classifying the orders of Hölder continuity of Brownian paths. We have till now shown that Brownian paths are Hölder continuous up to order $< 1/2$, and for orders $> 1/2$, they are not, but none of the previous proofs show what happens for order $1/2$. We will now show for order $1/2$, the Brownian paths are *not* Hölder continuous, and for this we need to introduce a new notion called *Lévy's modulus of continuity*.

6.3 Lévy's modulus of continuity

A modulus of continuity is a positive, continuous and strictly increasing function $\sigma : [0, \infty) \rightarrow [0, \infty)$ such that $\sigma(0) = 0$. This function can be used to derive some local smoothness properties of random or non-random function $f : [0, \infty) \rightarrow \mathbb{R}^d$ in terms of the following estimate:

$$|f(t) - f(s)| \leq \sigma(|s - t|), \text{ for all } s, t \geq 0. \quad (6.4)$$

By the above equation, it is not hard to see that it is enough to work with 1-dimensional functions. In the following theorem, we will show that $\sigma(t) := \sqrt{2t \log \frac{1}{t}}$ is the modulus of continuity for BM^1 . The positivity and continuity of the function is clear. Let us first prove an auxiliary result for the monotonicity of this function.

Theorem 6.3.1. *The function $\sigma(t) := \sqrt{2t \log \frac{1}{t}}$ is monotone increasing in the interval $(0, e^{-1})$. Moreover, for all $\kappa \in \mathbb{R}$ and $0 < h < \frac{1}{2}$, we have*

$$\sigma(2^\kappa h) \leq \sqrt{(|\kappa| + 1)2^\kappa} \sigma(h).$$

Proof. The first assertion is obtained by just squaring both sides and finding its first derivative to be positive in the interval $(0, e^{-1})$. The later is also easy to see by estimating $\frac{\sigma^2(2^\kappa h)}{\sigma^2(h)}$ in a canonical way. $\square$

Theorem 6.3.2 (Lévy 1937). *Let $(B_t)_{t \geq 0}$ be a BM^1 . Then,*

$$\mathbb{P} \left(\limsup_{h \rightarrow 0} \frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{2h \log \frac{1}{h}}} = 1 \right) = 1.$$

Corollary 6.3.3. *Almost all sample paths of one-dimensional Brownian motion are nowhere Hölder continuous of order $\alpha = \frac{1}{2}$.*

Proof. This proof is just a direct application of the the previous theorem. Note that

$$\frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{h}} = \frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{2h \log \frac{1}{h}}} \times \sqrt{2 \log \frac{1}{h}}.$$

Taking $\limsup_{h \rightarrow 0}$ on the right side, it tends to ∞ . It means that the B_t is nowhere Hölder continuous of order $1/2$ for $t \in [0, 1)$. This generalizes to the interval $[0, \infty)$ just by considering the BM $W(t) := B(t+h) - B(t)$ and applying the same proof as above. $\square$

Chapter 7

An overview of Itô's formula and stochastic differential equations

Stochastic calculus has enormous amount of applications in the field of mathematical finance. This chapter includes a short introduction to the basics of it.

By fundamental theorem of calculus, we know, that for "nice" functions f and g , we have

$$f(g(t)) - f(g(0)) = \int_0^t f'(g(s)) dg(s).$$

Now, let $(B_t)_{t \geq 0}$ be a BM¹ and $(\mathcal{F}_t)_{t \geq 0}$ be a right continuous admissible filtration. Then, if we try to apply our FTOC with the function g as our Brownian motion, then, we will get an extra corrector term, which arises because of the unbounded variation of Brownian paths. Let us see how such integrals are defined. We start off by some notations:

1. $\mathcal{M}_T^2$ for the family of $\mathcal{F}_t$ martingales $(M_t)_{0 \leq t \leq T} \subset L^2(\mathbb{P})$.
2. $M_T^{2,c}$ for the L^2 -martingales with a.s. continuous sample paths.
3. $L^2(\lambda_T \otimes \mathbb{P}) = L^2([0, T] \times \Omega, \lambda_T \otimes \mathbb{P})$, where λ_T is the Lebesgue measure on $[0, T]$.
4. L_T^2 is the set of all equivalence classes of processes from $L^2(\lambda_T \otimes \mathbb{P})$ which have an *progressively measurable* representative. (A process X is said to be progressively measurable if for each t , X_t is $(\mathcal{B}[0, t] \otimes \mathcal{F}_t)$ -measurable.)

Definition 7.0.1 (Simple processes). A real-valued stochastic process $(f(t, \cdot))_{t \in [0, T]}$ of the form

$$f(t, \omega) = \sum_{j=1}^n \phi_{j-1}(\omega) \mathbf{1}_{[s_{j-1}, s_j]}(t),$$

where $n \geq 1$, $0 = s_0 \leq s_1 \leq \dots \leq s_n \leq T$, and $\phi_j \in L^\infty(\mathcal{F}_{s_j})$ are bounded $\mathcal{F}_{s_j}$ measurable random variables, $j=0, 1, \dots, n-1$, is called a (right continuous) *simple process*. We denote $\mathcal{S}_T$ for the family of all simple processes on $[0, T]$.

Definition 7.0.2 (Itô integral for simple processes). Let $M \in M_T^{2,c}$ be a continuous L^2 martingale and $f \in \mathcal{S}_T$. Then

$$f \bullet M_t := \int_0^t f(s) dM_s := \sum_{j=1}^n f(s_{j-1})(M(s_j) - M(s_{j-1})).$$

Now, this above definition is used in the same way as Riemann-Stieltjes integral is defined, by the method of approximation, to define the Itô integral for a Brownian motion.

Definition 7.0.3 (Itô integral). Let $(B_t)_{t \geq 0}$ be a BM^1 and let f in L_T^2 . Then, the Itô integral is defined as

$$f \bullet B_t = \int_0^t f(s) dB_s := L^2 - \lim_{n \rightarrow \infty} f_n \bullet B_t, \quad 0 \leq t \leq T,$$

where $(f_n)_{n \geq 1} \subset \mathcal{S}_T$ is any sequence approximating $f \in L_T^2$ in $L^2(\lambda_T \otimes \mathbb{P})$.

The fact that the limit in the definition exists is easy to see by applying Cauchy criterion as L^2 is a Hilbert space. Now, the famous Itô's formula which forms the basis of most of stochastic calculus is given by the following theorem.

Theorem 7.0.4 (Itô, 1942). Let $(B_t)_{t \geq 0}$ be a BM^1 and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^2 function. Then, for all $t \geq 0$, we have a.s.

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad (7.1)$$

It is given by another theorem of Itô that the above time-dependent formula holds for all $f \in C^{1,2}([0, \infty), \mathbb{R}^m)$.

7.1 Itô's processes

If $T \in [0, \infty]$, then, stochastic processes of the form

$$X_t - X_0 = \int_0^t \mu(X_s, s) ds + \int_0^t \sigma(X_s, s) dB_s, \quad t < T \quad (7.2)$$

are called Itô's processes. If the integrands satisfy $\sigma \in L_{T,loc}^2$ and for almost all ω , $\mu(s, \omega) \in L^1([0, T], ds)$, then, the above equation makes sense. This equation characterizes the behaviour of a continuous time stochastic process X_t as a sum of an ordinary Lebesgue integral and an Itô integral. Heuristically, an Itô process is locally a Brownian motion, in the sense that in a small time interval of length δ , the stochastic process X_t changes its value by an amount that is normally distributed with expectation $\mu(X_s, s)\delta$ and variance $\sigma^2(X_s, s)\delta$ and is independent of the past behaviour of the process. This is so because the increments of a Wiener process (standard BM) are independent and normally distributed. The function μ is called the *drift* co-efficient and σ is called the *diffusion* co-efficient. Such a stochastic process X_t which satisfies this equation is also called a *diffusion process*, and it satisfies the Markov property.

This integral can also be generalized to dimensions higher than $d = 1$, in which case, the integrals are defined coordinate wise. The equation 7.1 also has another generalization when the function f is also dependent upon time, i.e. $f := f(t, B_t)$. Then the generalized Itô's formula is shown in the next theorem.

Theorem 7.1.1. Let $(B_t)_{t \geq 0}$ be a BM^d and X_t given by 7.2 is a m -dimensional Itô process. Then, for every C^2 function $f : [0, \infty) \times \mathbb{R}^m \rightarrow \mathbb{R}$,

$$\begin{aligned} df(t, X_t) = & \partial_t f(t, X_t) dt + \sum_{j=1}^m \partial_j f(t, X_t) \mu_j(t) dt + \sum_{j=1}^m \sum_{k=1}^d \partial_j f(t, X_t) \sigma_{jk}(t) dB_t^k \\ & + \frac{1}{2} \sum_{i,j=1}^m \partial_i \partial_j f(t, X_t) \sum_{k=1}^d \sigma_{ik}(t) \sigma_{jk}(t) dt. \end{aligned}$$

7.2 Stochastic differential equations

Differentiating both sides of equation 7.2, we obtain the general equation of SDEs, which is

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t. \quad (7.3)$$

Now, we give the precise definition of solution of the differential equation 7.3.

Definition 7.2.1. Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^d , and $\mu : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ be measurable functions. Then, a solution for the SDE 7.3 with initial condition $X_0 = \xi$ is a progressively measurable stochastic process $(X_t)_{t \geq 0}$, $X_t = (X_{t_1}, \dots, X_{t_n}) \in \mathbb{R}^n$ such that the following integral is satisfied:

$$X_t = \xi + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s,$$

i.e. for all $j = 1, 2, \dots, n$, and $t \geq 0$,

$$X_t^j = \xi^j + \int_0^t \mu_j(s, X_s) ds + \sum_{k=1}^d \int_0^t \sigma_{jk}(s, X_s) dB_s^k.$$

(It is implicit in the definition that all the stochastic integrals make sense.)

Let us now look at some examples of SDE. To keep things simple, we only consider the cases $d = n = 1$. Throughout the next section, we will consider that all the integrals into play exists.

7.3 Existence and uniqueness of solutions

One might have already observed that if $\sigma = 0$, then, our 7.3 becomes an ODE. So, the theorems of this section will obviously have the same flavour as we found while studying the existence and uniqueness results of ODE. So, it seems reasonable to require the following Lipschitz condition:

$$\sum_{j=1}^n |\mu_j(t, x) - \mu_j(t, y)|^2 + \sum_{j=1}^n \sum_{k=1}^d |\sigma_{jk}(t, x) - \sigma_{jk}(t, y)|^2 \leq L^2 |x - y|^2, \quad (7.4)$$

for all $x, y \in \mathbb{R}^n$, $t \in [0, T]$ and a global Lipschitz constant $L = L_T$. Also, if $y = 0$, the global Lipschitz condition implies the following linear growth condition

$$\sum_{j=1}^n |\mu_j(t, x)|^2 + \sum_{j=1}^n \sum_{k=1}^d |\sigma_{jk}(t, x)|^2 \leq M^2 (1 + |x|)^2, \quad (7.5)$$

for all $x \in \mathbb{R}^n$ and with some constant $M = M_T$ which can be estimated as $M^2 \geq 2L^2 + 2 \sum_{j=1}^n \sup_{t \leq T} |\mu_j(t, 0)|^2 + 2 \sum_{j=1}^n \sum_{k=1}^d \sup_{t \leq T} |\sigma_{jk}(t, 0)|^2$.

Theorem 7.3.1 (Existence). *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^d and assume that the coefficients $\mu : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ satisfy the Lipschitz condition 7.4 for all $x, y \in \mathbb{R}^n$ with a global Lipschitz constant $L = L_T$ for $T > 0$. For every $\mathcal{F}_0$ measurable initial condition $X_0 = \xi \in L^2(\mathcal{P})$, there exists a unique solution $(X_t)_{t \geq 0}$, and this solution satisfies*

$$\mathbb{E} \left[\sup_{s \leq T} |X_s|^2 \right] \leq \kappa_T \mathbb{E} [1 + |\xi|^2], \text{ for all } T > 0.$$

The uniqueness result is a corollary to the following stability condition:

Theorem 7.3.2 (Stability). *Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^d , and assume that the coefficients $\mu : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ satisfy the Lipschitz condition 7.4 for all $x, y \in \mathbb{R}^n$ with a global Lipschitz constant $L = L_T$ for $T > 0$. If two solutions $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ are any two solutions of the SDE 7.3 with $\mathcal{F}_0$ measurable initial conditions $X_0 = \xi \in L^2(\mathcal{P})$ and $Y_0 = \eta \in L^2(\mathcal{P})$, respectively, then*

$$\mathbb{E} \left[\sup_{t \leq T} |X_t - Y_t|^2 \right] \leq 3e^{3L^2(T+4)T} \mathbb{E} [|\xi - \eta|^2].$$

Clearly, if we consider the initial conditions of X_t and Y_t to be equal say, $X_0 = Y_0 = \xi$, then, from the above estimate, $\mathbb{P}(X_t = Y_t, \text{ for all } t \in [0, n]) = 1$, for all $n \geq 1$, hence it also holds for all $t \geq 0$.

Remark 7.3.3. *Geometric Brownian motion* comes as a solution to the following SDE and is the one of the most celebrated models used in finance, for example it is heavily used in modelling stock prices:

$$X_t = X_0 + b \int_0^t X_s ds + \sigma \int_0^t X_s dB_s, \quad X_0 = x_0 \in \mathbb{R},$$

where $b \in \mathbb{R}$ and $\sigma > 0$ are constants. The solution is given by

$$X_t = x_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right], \quad t \geq 0.$$

Itô's formula can be used directly to the above process to verify that it is indeed the solution of the aforementioned SDE.

Chapter 8

Simulation of Brownian motion

Till yet, we have focused on studying all the theoretical aspects of a BM, but in this chapter we will learn how to simulate a Brownian motion. Simulation is the process of generating synthetic data that closely mimics the properties and characteristics of real-world data.

For iid random variables $X_1, X_2, \dots, X_n$ with common distribution F , for any $\omega \in \Omega$, the tuple $(X_1(\omega), X_2(\omega), \dots, X_n(\omega))$ is called a *true sample* of F , whereas any tuple of values $(x_1, x_2, \dots, x_n)$, which is indistinguishable from a true sample of F by statistical inference is called a *sample* of F , and any element of such a tuple is called a *sample value* of F , which is denoted by

$$x_j \stackrel{s}{\sim} F.$$

Nevertheless, in simulations one always treats a sample as if it were a true sample.

8.1 Transformation of sample distribution

Let us begin with learning how we can convert a sample from one particular distribution into a sample of some other distribution. The following is the most basic algorithm used for this purpose. This is the

Algorithm 1 Inverse transform method

Require: F a distribution function

$$\begin{aligned} u &\stackrel{s}{\sim} U_{(0,1)} \\ x &\leftarrow F^{-1}(u) \end{aligned}$$

Then $x \stackrel{s}{\sim} F$.

sample generating version of the algorithm that one might have already used before while generating a random variable X with a certain distribution F , by considering a uniform random variable, say U , over any interval $[a, b]$ and then taking the generalized inverse $F^{-1}(U) \sim F$.

Example 8.1.1 (Exponential distribution Exp_λ). The distribution function of Exp_λ with parameter $\lambda > 0$ is $F(x) = 1 - e^{-\lambda x}$, $x \geq 0$ and

$$F^{-1}(y) = -\frac{1}{\lambda} \log(1 - y).$$

Now, considering $u \stackrel{s}{\sim} U_{(0,1)}$, the fact $1 - u \stackrel{s}{\sim} U_{(0,1)}$ yields with the inverse transform method

$$-\frac{1}{\lambda} \log(u) \stackrel{s}{\sim} \text{Exp}_\lambda.$$

The next algorithm, known as acceptance-rejection method, comes super handy when other methods like the inverse transform method fails. This algorithm is used in cases where generating samples from a particular distribution, say F with a density function $f(x)$ is fairly difficult. The basic idea is to find an alternative distribution G with density function $g(x)$ such that the function $f(x)$ is close to $g(x)$, that is, for a constant c , $\sup_x \{ \frac{f(x)}{g(x)} \} \leq c$, and in practice, c is desired to be as close to 1.

Algorithm 2 Acceptance-rejection method

Require: G distribution function as above

$y \stackrel{s}{\sim} G$

$u \stackrel{s}{\sim} U_{(0,1)}$

if $u \leq \frac{f(y)}{cg(y)}$ **then**

$x \leftarrow y$

▷ "accept"

else restart with 1

▷ "reject"

end if

Proof. The proof of this algorithm is a fairly routine calculation of conditional distribution. It can easily verified that

$$\mathbb{P} \left(Y \leq y \middle| U \leq \frac{f(Y)}{cg(Y)} \right) = F(y),$$

by using the fact that $\mathbb{P} \left(U \leq \frac{f(Y)}{cg(Y)} \right) = \frac{1}{c}$, which is also not difficult to calculate. $\square$

Example 8.1.2. The above algorithm can be used for sampling from the one sided normal distribution

$$f(x) = 2 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \mathbb{1}_{[0,\infty)}(x),$$

and the Exp_1 distribution $g(x) = e^{-x} \mathbb{1}_{[0,\infty)}$ with the smallest possible value of c given by

$$c = \sup_x \frac{f(x)}{g(x)} = \sqrt{\frac{2e}{\pi}} \approx 1.315.$$

Following is another very useful algorithm used to generate samples from a distribution (which if done directly, might be a slow process) which can be written as a weighted sum or integral of other distributions from which sampling might be easier.

Algorithm 3 Composition method

Require: F be a distribution which admits density function f of the form

$$f(x) = \int g_y(x) \mu(dy),$$

where, μ is a probability measure, and g_y is for each y a density.

$y \stackrel{s}{\sim} \mu$

$x \stackrel{s}{\sim} g_y$

Then, $x \stackrel{s}{\sim} F$.

Proof. Let $Y \sim \mu$, and then conditioned on $\{Y = y\}$, let $X \sim g_y$. Then, by the law of total probability, and Fubini's theorem,

$$\mathbb{P}(X \leq x) = \int \mathbb{P}(X \leq x | Y = y) \mu(dy) = \int \int_{-\infty}^x g_y(z) dz \mu(dy) = \int_{-\infty}^x f(z) dz = F(x).$$

Thus, we have $x \sim F$. $\square$

Example 8.1.3. Let us see a very easy application of the above algorithm. In the example of the previous algorithm, we have seen how we can generate samples from a one-sided distribution normal distribution. Using these samples, we can generate samples from a standard normal distribution by composition algorithm.

The density of a standard normal distribution can be decomposed as follows:

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = \frac{1}{2} \times \frac{2}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \mathbf{1}_{(-\infty, 0]} + \frac{1}{2} \times \frac{2}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \mathbf{1}_{(0, \infty)} \\ &= \frac{1}{2} g_{-1}(x) + \frac{1}{2} g_1(x). \end{aligned}$$

8.2 Brownian motion

In this section, we present many algorithmic implementation of the constructions of Brownian motion that we have seen in previous sections. Note that since we can only generate finitely many samples in a finite time, thus we will have to linearly interpolate between them to simulate a continuous time process.

Our goal is to simulate a continuous Brownian path. We first begin by the most basic approach to this end.

8.2.1 Independent increments

We simulate a Brownian path by stationary and independent increments. Let $(B_t)_{t \geq 0}$ be a Brownian motion. At the end of the algorithm, we get $(b_0, b_{t_1}, \dots, b_{t_n}) \sim (B_0, B_{t_1}, \dots, B_{t_n})$, as required.

Algorithm 4 Independent increments

Require: $n \geq 0, 0 = t_0 < t_1 < \dots < t_n$

```

 $b_0 \leftarrow 0$ 
 $j \leftarrow 1$ 
while  $j \leq n$  do
   $y \sim N(0, t_j - t_{j-1})$ 
   $b_{t_j} \leftarrow b_{t_{j-1}} + y$ 
   $j \leftarrow j + 1$ 
end while

```

8.2.2 Interpolation via Levy's argument

Using Algorithm 4, we may obtain some samples from a Brownian path. However, if we refine it then we will get a different path. Since our goal is to refine one Brownian path repeatedly, so we may use the following algorithm to interpolate while fixing the Brownian path.

Algorithm 5 Interpolation via Levy's argument

Require: $s_0 < s < s_1, (b_{s_0}, b_{s_1}) \sim (B_{s_0}, B_{s_1})$

$$b_s \sim N\left(\frac{(s_1 - s)b_{s_0} + (s - s_0)b_{s_1}}{s_1 - s_0}, \frac{(s - s_0)(s_1 - s)}{s_1 - s_0}\right)$$

At the end of the algorithm, we claim that we will get

$$(b_{s_0}, b_s, b_{s_1}) \sim (B_{s_0}, B_s, B_{s_1})$$

under the condition that $B_{s_0} = b_{s_0}$ and $B_{s_1} = b_{s_1}$. Indeed, this is what we saw in §2.1.2.

8.2.3 Levy-Ciesielski

Using Algorithm 4, we may initialise a Brownian path. Using Algorithm 5, we may obtain a refinement of the same path. We may refine only at the dyadic points to get the Levy-Ciesielski representation.

Algorithm 6 Interpolation via Levy's argument

Require: $J \geq 1$

```

 $b_0 \leftarrow 0$ 
 $j \leftarrow 0, l \leftarrow 0$ 
while  $j \leq J - 1$  do
  while  $l \leq 2^j - 1$  do
     $y \sim N(0, 1)$ 
     $b_{\frac{2l+1}{2^{j+1}}} \leftarrow \frac{1}{2} \left( b_{\frac{l}{2^j}} + b_{\frac{l+1}{2^j}} \right) + 2^{-(\frac{j}{2}+1)} y$ 
     $l \leftarrow l + 1$ 
  end while
   $j \leftarrow j + 1$ 
end while

```

At the end of the algorithm, we get that

$$(b_0, b_{1/2^J}, b_{2/2^J}, \dots, b_1) \sim (B_0, B_{1/2^J}, B_{2/2^J}, \dots, B_1).$$

8.2.4 Donsker's invariance principle

We can also interpolate using Theorem 2.4.1. At the end of the algorithm, by Theorem 2.4.1, it follows

Algorithm 7 Interpolation via Levy's argument

Require: $n \geq 1$

```

 $b_0 \leftarrow 0$ 
 $u_j \sim U_{\{-1,1\}}$  for each  $j = 1, \dots, n$ 
 $k \leftarrow 0$ 
while  $k \leq n$  do
   $b_{k/n} \leftarrow \frac{1}{\sqrt{n}} u_k + b_{\frac{k-1}{n}}$ 
   $k \leftarrow k + 1$ 
end while

```

that

$$(b_0, b_{1/n}, b_{2/n}, \dots, b_1) \sim (B_0, B_{1/n}, B_{2/n}, \dots, B_1),$$

as required.